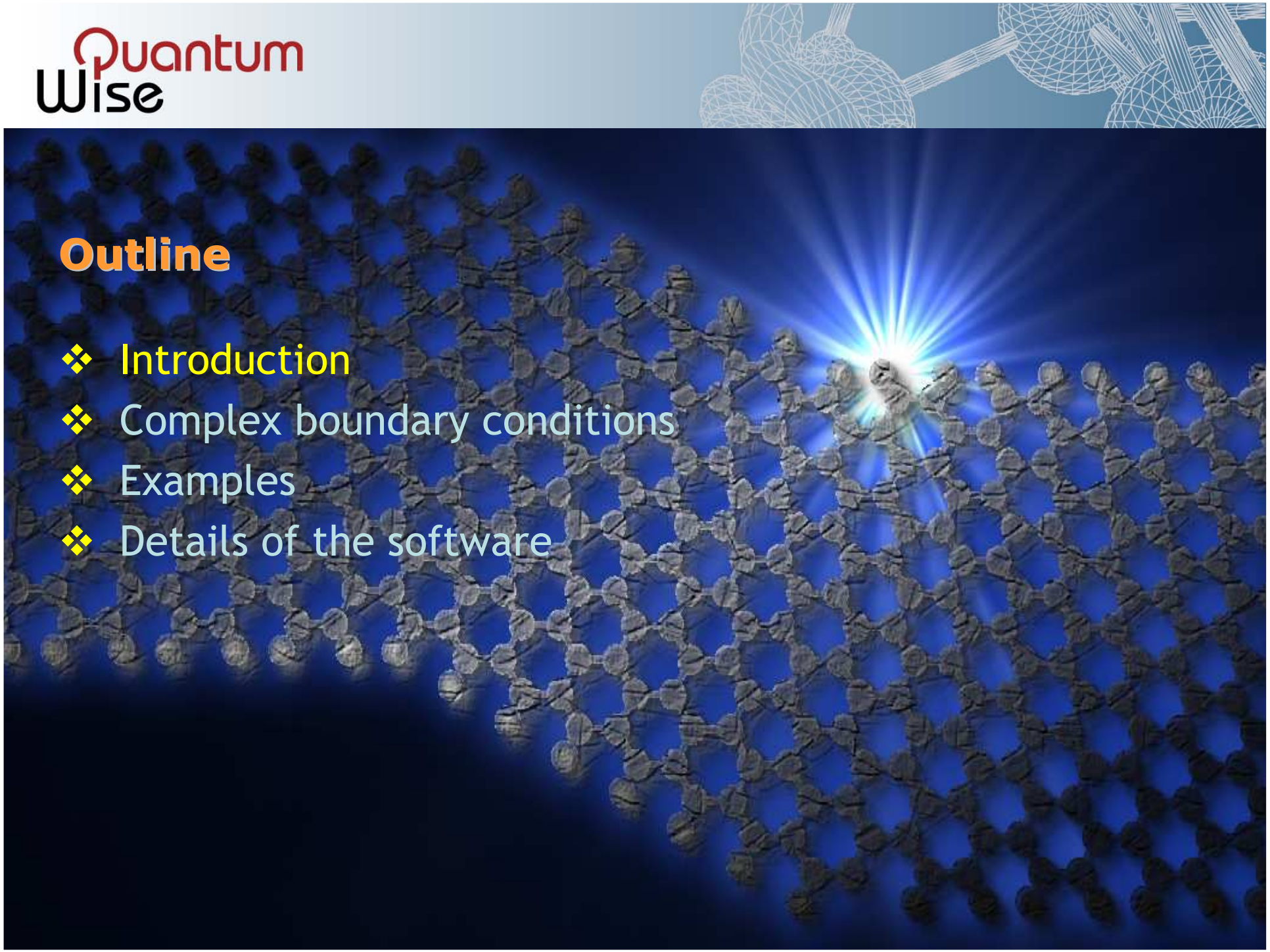
Atomistix ToolKit
Modeling of nanoelectronic devices

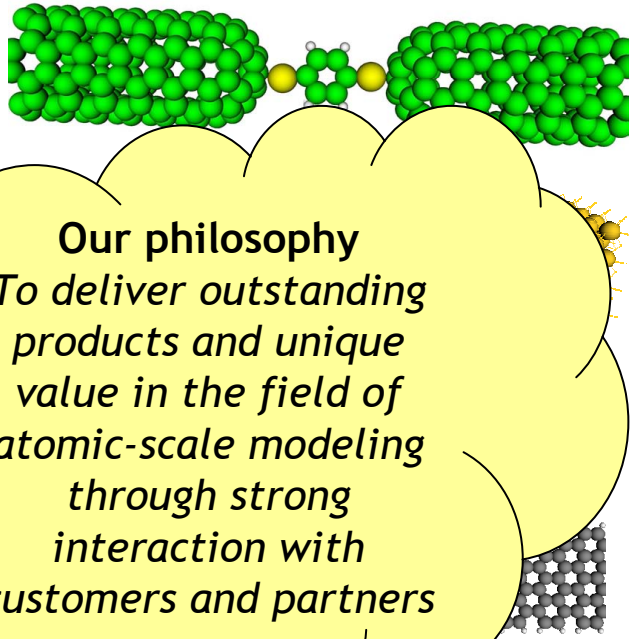
Anders Blom
QuantumWise A/S
www.quantumwise.com

QuantumWise

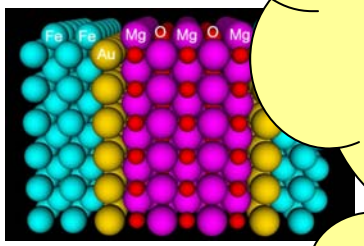

Outline
Introduction
Complex boundary conditions
Examples
Details of the software

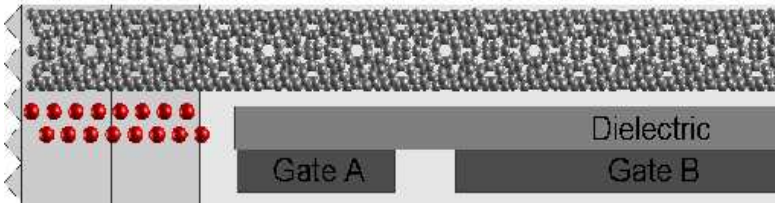
Atomistix ToolKit, introduction

- Simulations of electronic transport in nanoscale devices
- Unique functionality
  Focus on large-scale systems
- Modern software platform, combining powerful scripting with a graphical user interface

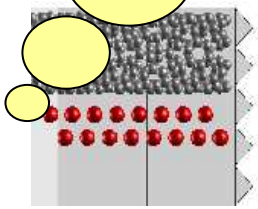


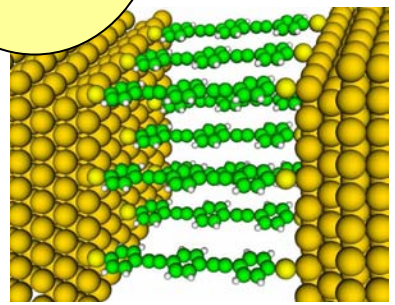
Our philosophy
To deliver outstanding products and unique value in the field of atomic-scale modeling through strong interaction with customers and partners





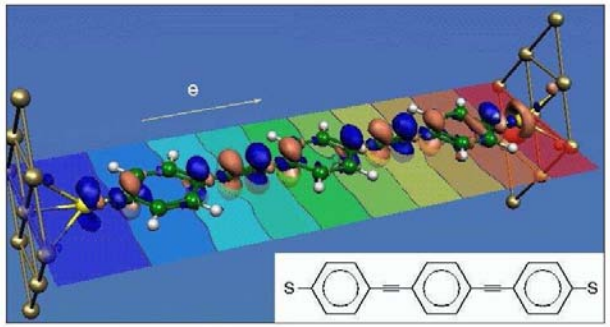




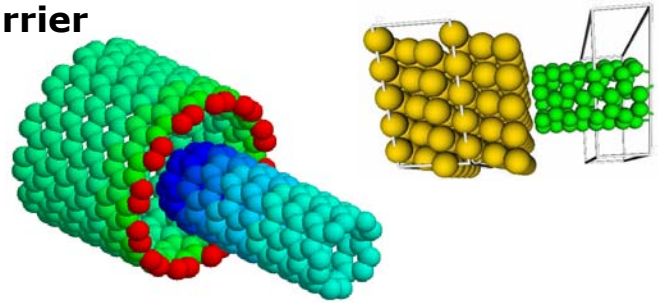


Application areas (1/2)

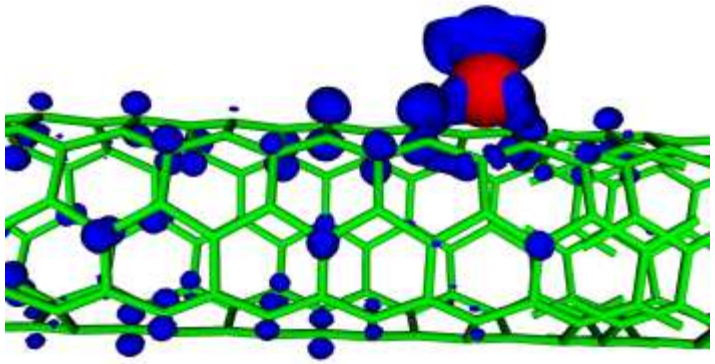
Molecular junctions and tunneling devices (rectification, NDR)



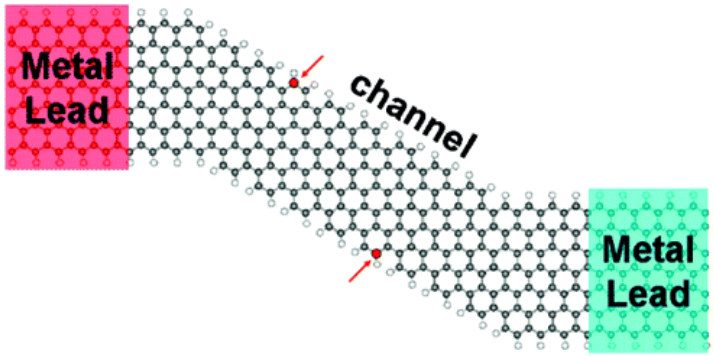
Contact resistance and capacitance of metal-nanotube and nanotube-nanotube contacts; Schottky barrier



Functionalized carbon nanotubes

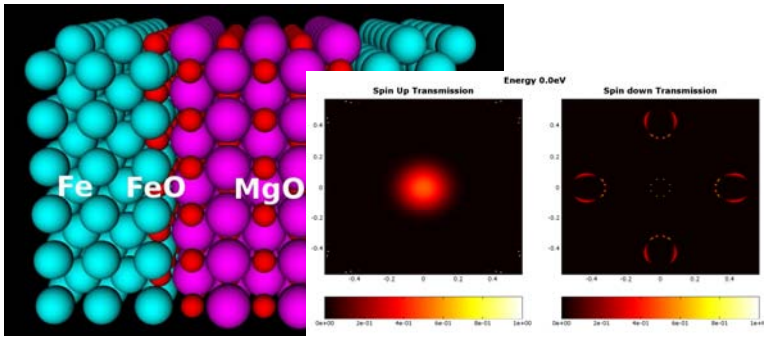


Graphene devices

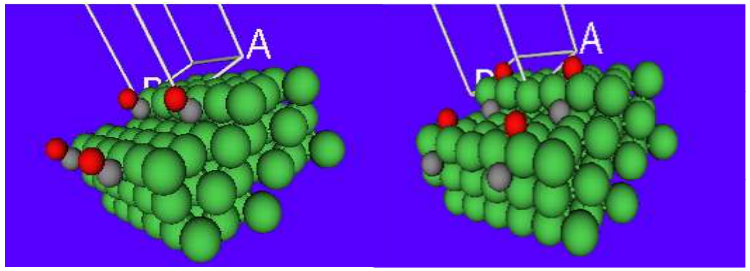


Application areas (2/2)

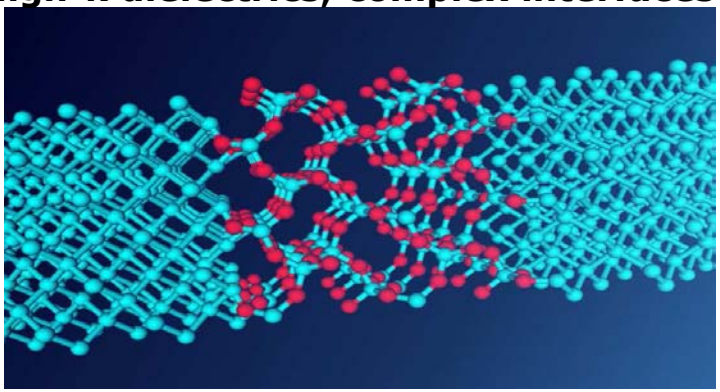
Spin-dependent transport across crystalline magnetotunnel junctions



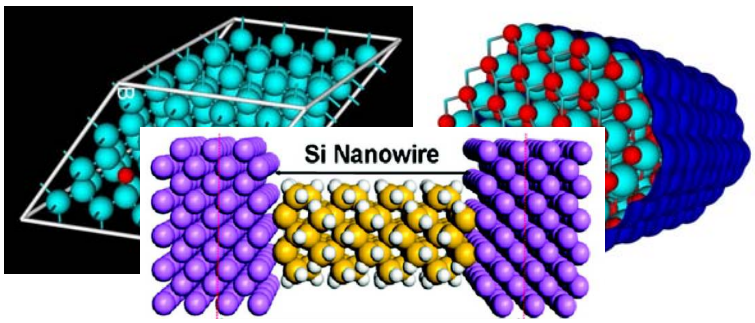
Reaction paths on surfaces for catalysis



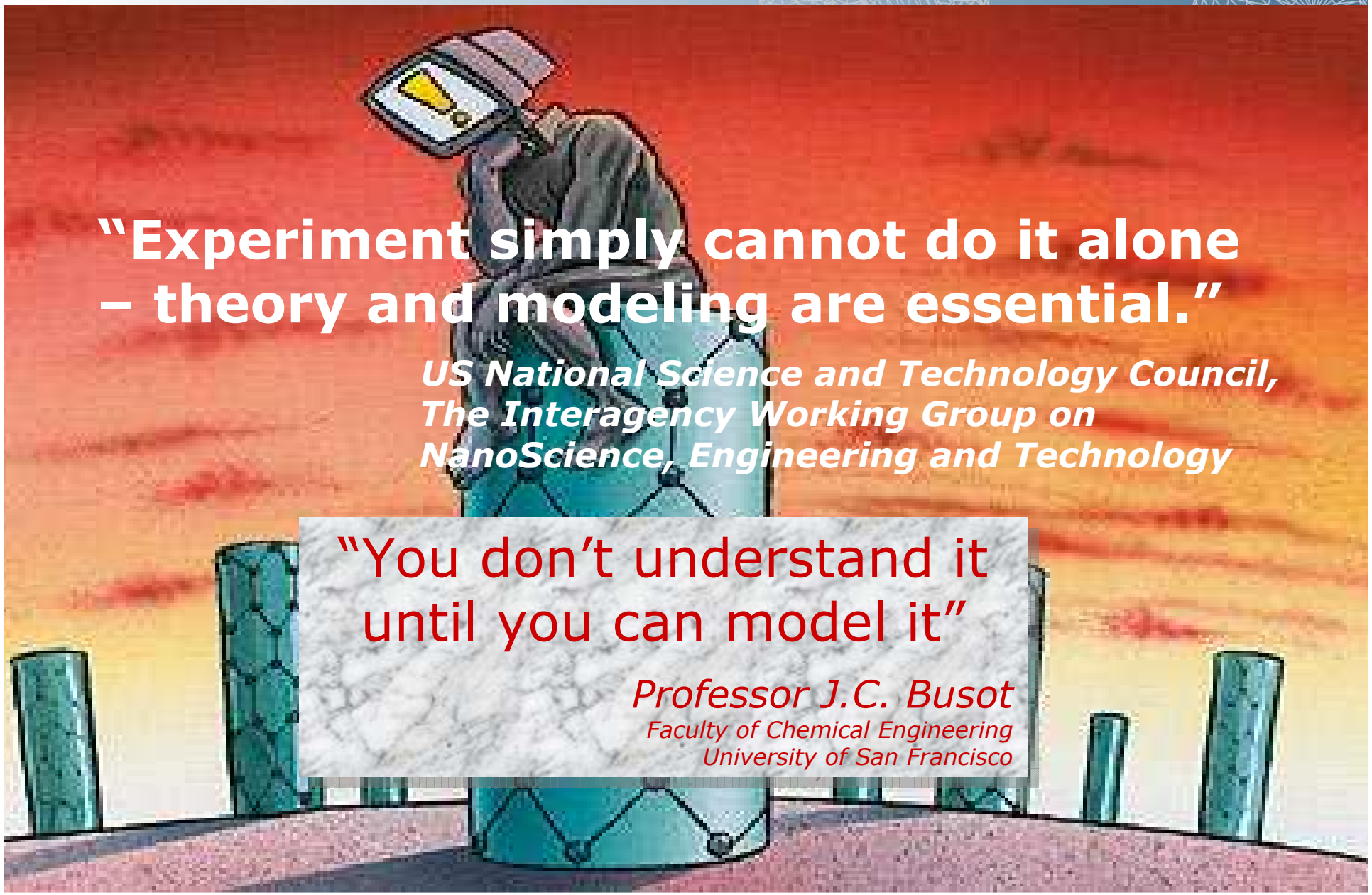
Leakage currents in MOS structures High-k dielectrics, complex interfaces



Defect states in semiconductors, nanowires, nanotubes



Quantum Wise



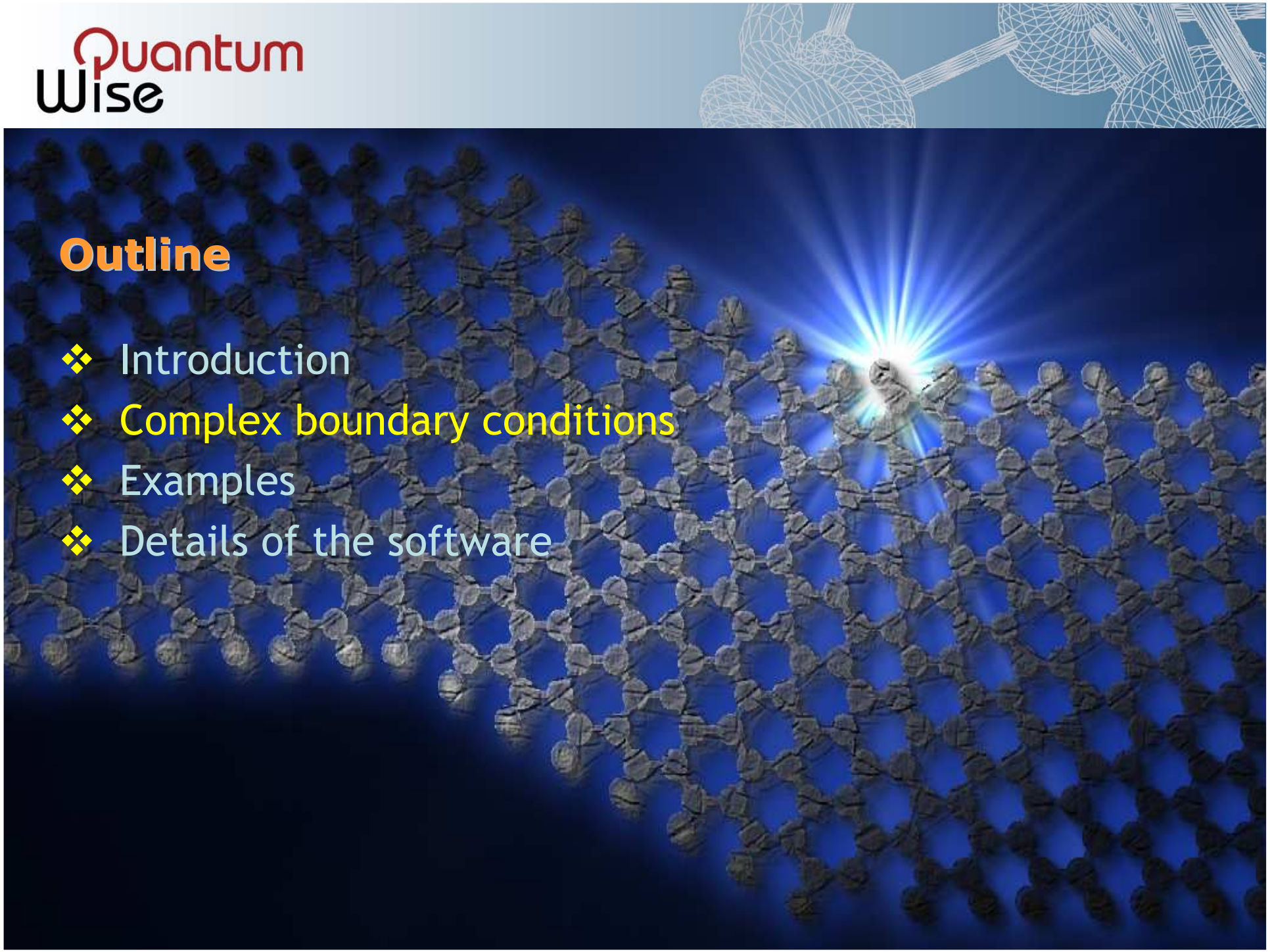
"Experiment simply cannot do it alone – theory and modeling are essential."

US National Science and Technology Council, The Interagency Working Group on NanoScience, Engineering and Technology

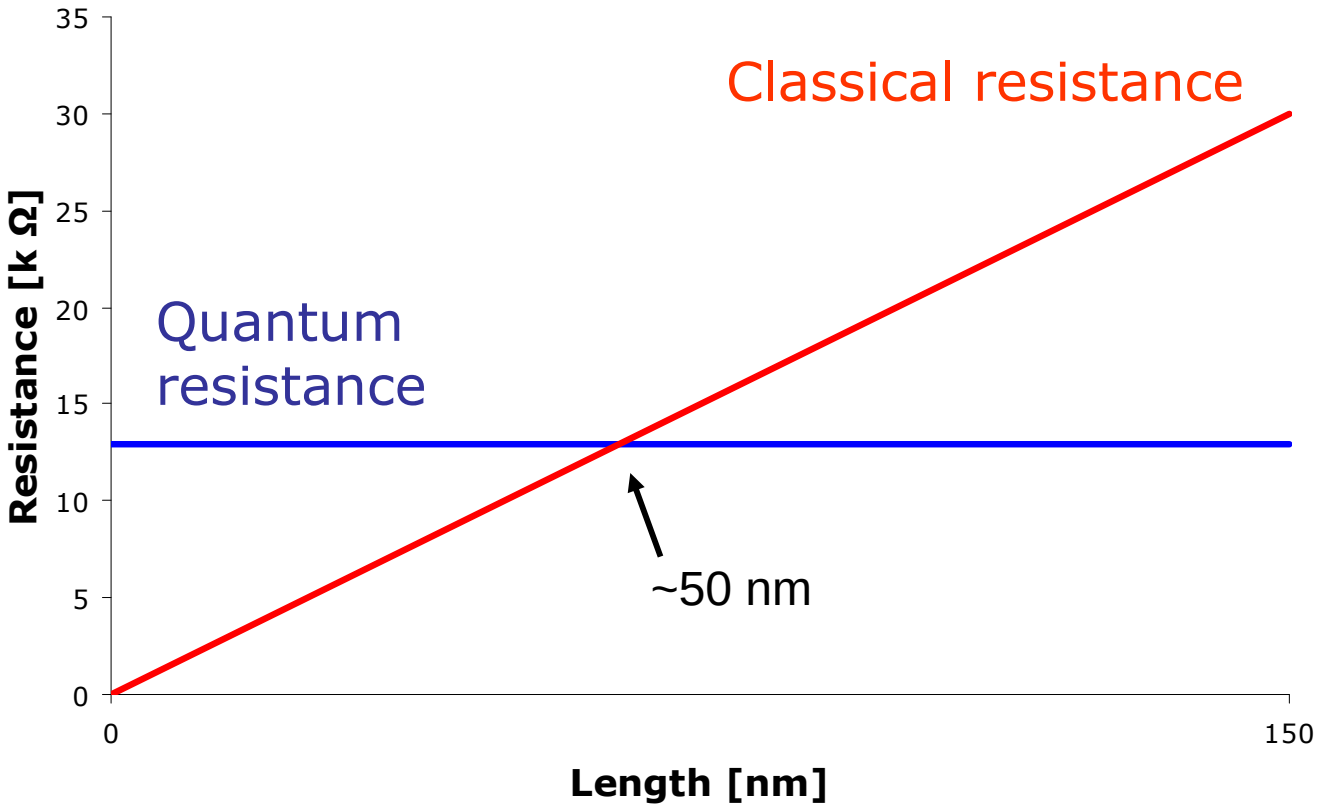
"You don't understand it until you can model it"

Professor J.C. Busot

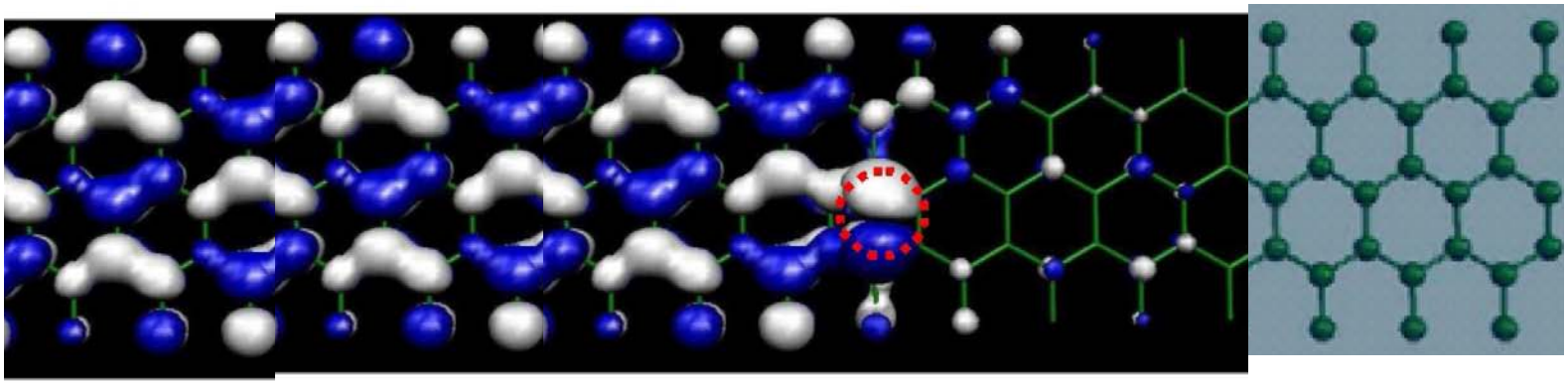
Faculty of Chemical Engineering
University of San Francisco


Outline
Introduction
Complex boundary conditions
Examples
Details of the software

Reaching the quantum limit



Current mechanism



Propagation of wave function

Ballistic, coherent transport

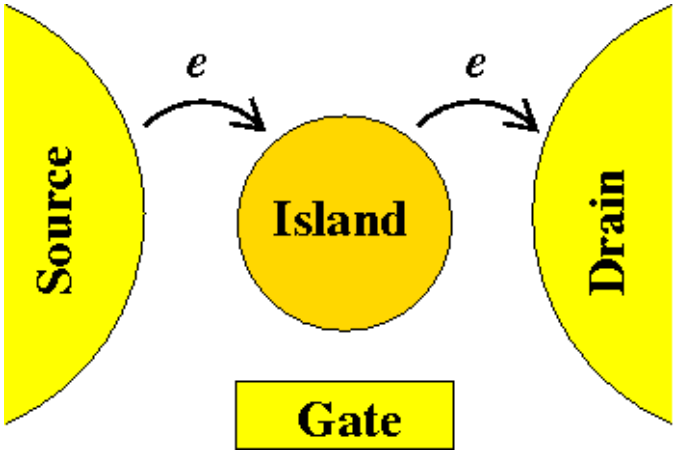
Calculation of transmission amplitudes **t**

$$I = \frac{2e^2}{h} \int dE (f_R(E) - f_L(E)) T_{\text{tot}}(E)$$

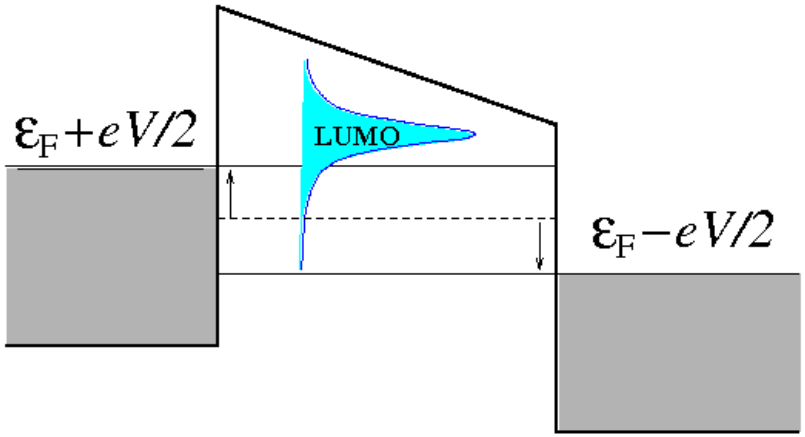
$T_{\text{tot}}(E)$: Total Quantum Transmission

$$T_{\text{tot}}(E) = \text{Tr}[\mathbf{t}^\dagger \mathbf{t}](E)$$

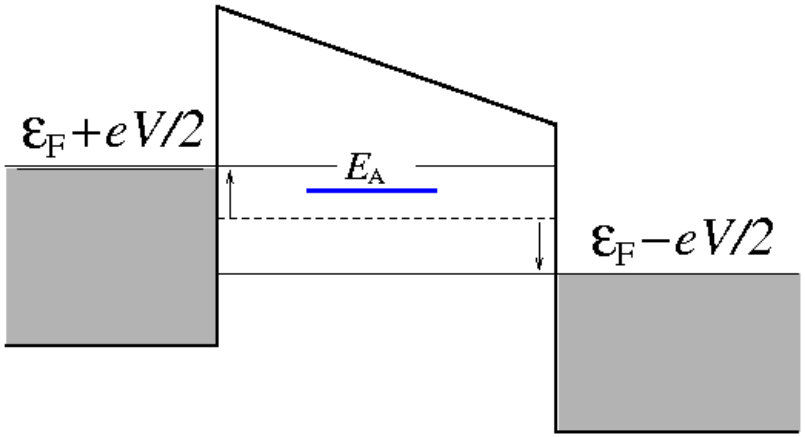
Transport regimes



Coherent Tunneling Regime



Sequential Tunneling Regime

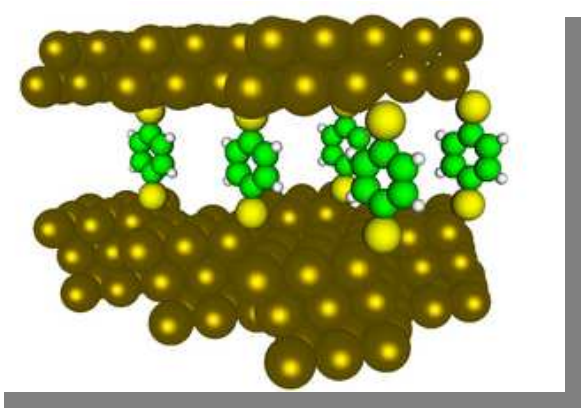


Essential points of atomistic device modeling

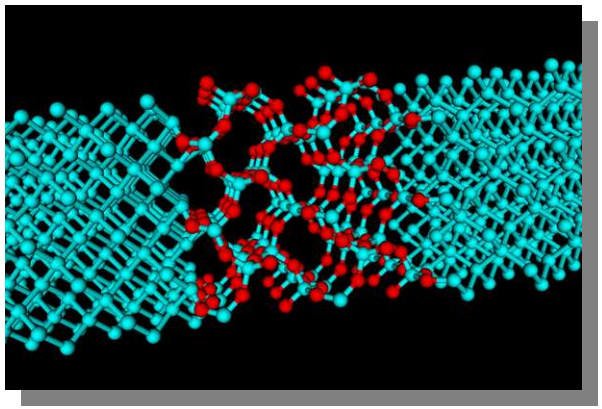
- Atomistic, quantum models
- Accurate electronic structure
  - Tight-binding works – when it works…
  - DFT most transferable, but heavy. Semiconductors?
  - Semi-empirical methods
- Describe voltage drop
  - Self-consistent response to electrostatic environment
- Proper boundary conditions
  - Transport under finite bias -> NEGF
  - Gates
- Transmission spectrum usually bias-dependent
  - Non-linear response

Complex boundary conditions

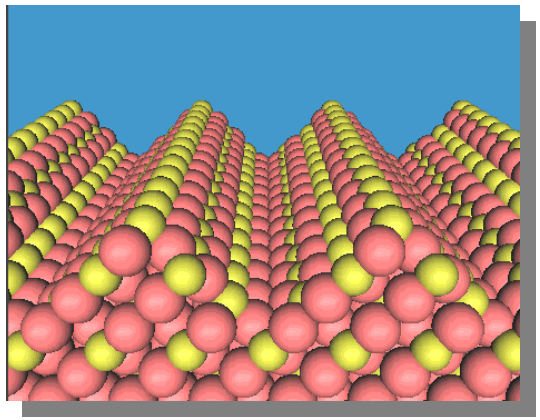
The main application areas in nanotechnology are related to effects occurring at



Junctions

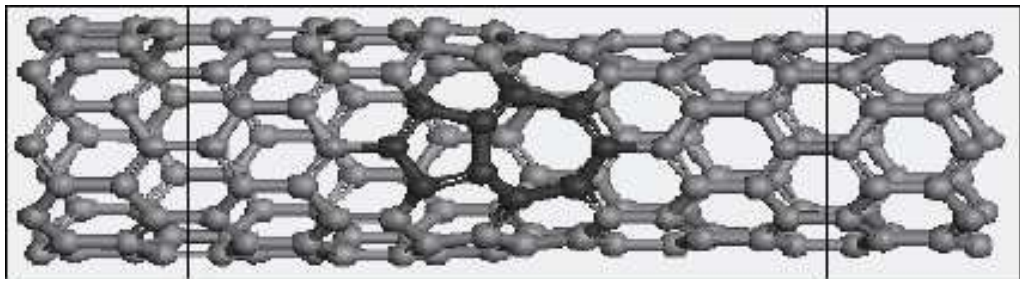


Interfaces



Surfaces

It is critical to be able to accurately model systems with complex boundary conditions from quantum theory

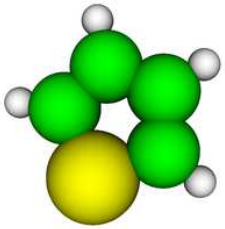


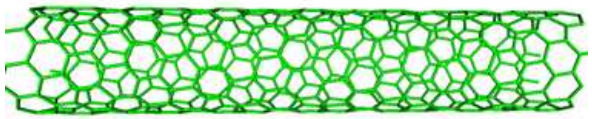
Electronic transport in nanostructures

Traditional quantum-based software can model either

isolated molecules or periodic systems

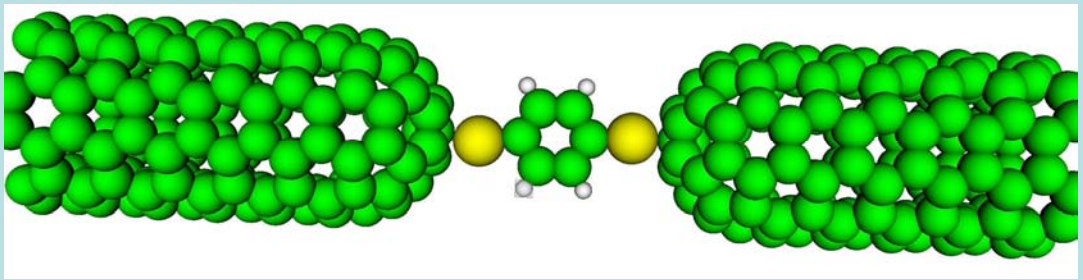
Gaussian
DMOL
TurboMole

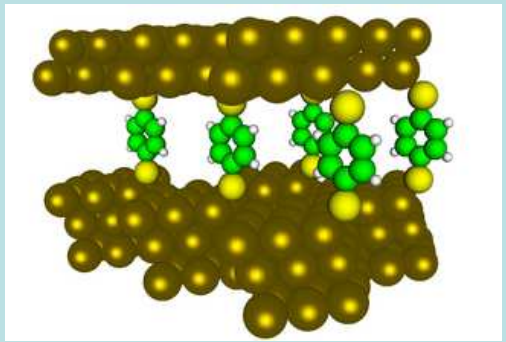




VASP
CASTEP
DMol3
Wien2k

Devices are more complex; need to model nanostructures combining molecules with periodic systems and macroscopic elements

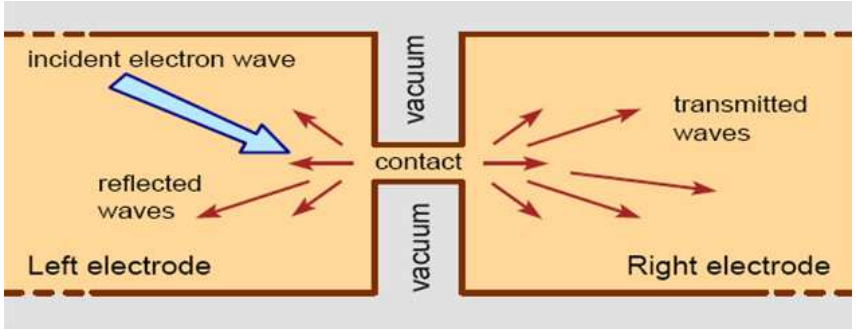


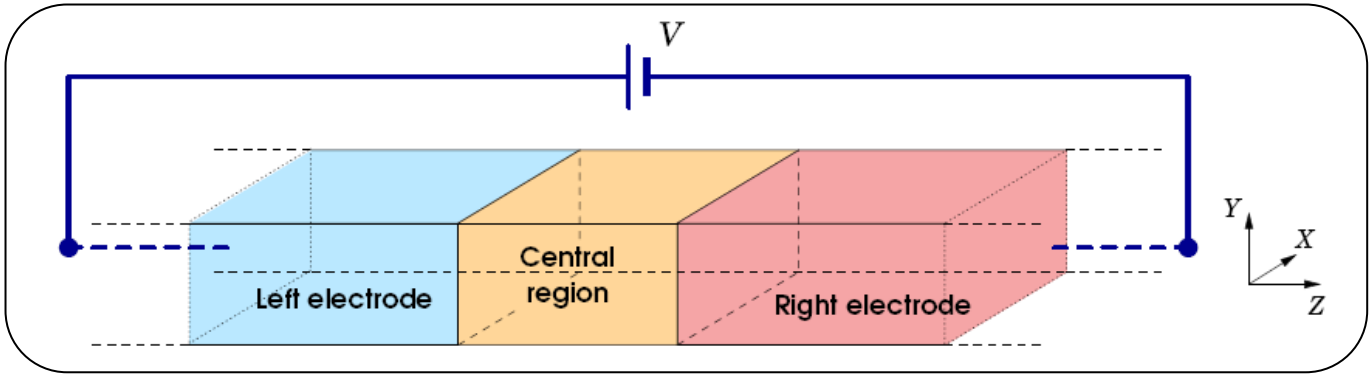


This enables simulations of electrical properties of complex nanoscale components and systems like transistors

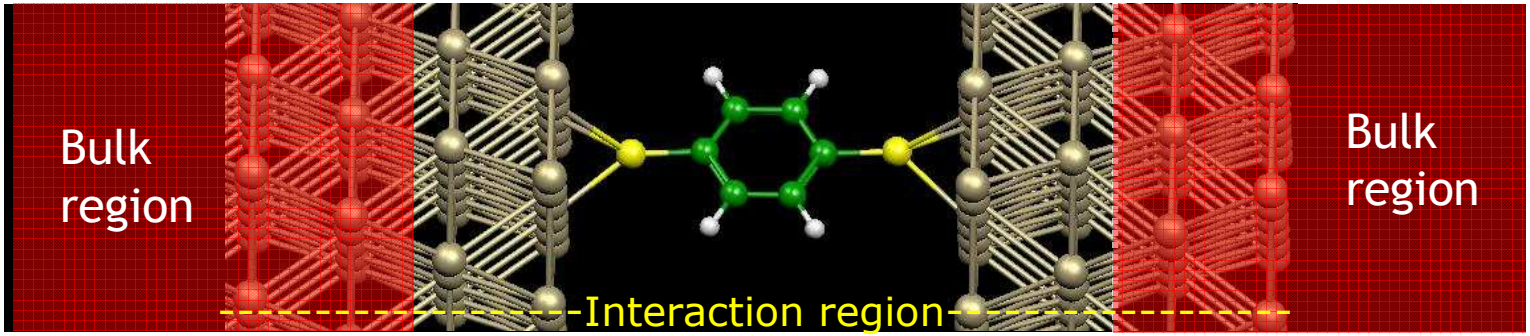
Transport boundary conditions

- Scattering boundary conditions apply in the transport direction
  - Electrons flow through the system under the influence of an applied voltage bias
  - Non-equilibrium electron distribution described by NEGF
  - Electric current is due to ballistic, coherent tunneling, computed using the Landauer formalism





Two-probe methodology



Electronic structure

- Density-functional theory *or* extended Hückel
- LCAO, numerical orbitals with finite range
- DFT uses norm-conserving pseudopotentials
- Extended Hückel has Hartree term for SCF

D

H

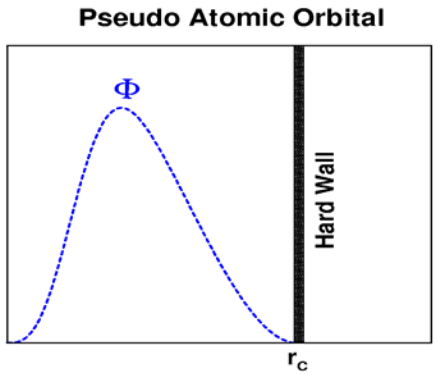
Transport

- Exact description of semi-infinite electrodes using self-energies
- Non-equilibrium electron distribution using NEGF
- Calculation of electron current by Landauer formalism

M. Brandbyge, J.-L. Mozos, P. Ordejón, J. Taylor, K. Stokbro, Physical Review **B 65**, 165401 (2002)

DFT model

- SIESTA method
  - Own implementation
- Linear combination of numerical atomic orbitals (LCAO)
- Finite range
  - Sparse matrices (+)
  - Non-variational (–)
- Norm-conserving Troullier-Martins pseudopotentials
- LDA/GGA



J. M. Soler, E. Artacho, J. D. Gale, A. García, J. Junquera, P. Ordejón, and D. Sánchez-Portal, *J. Phys. Condens. Matter* **14**, 2745 (2002)

Extended Hückel model

- Self-consistent, robust model, popular in molecular physics
- Builds on published and well-tested algorithms
- Faster than DFT
- Smaller basis sets -> larger systems
- Tunable via tailor-made parameters for specific problems
  - Can be more accurate than DFT – if you find good parameters
  - Less transferable than DFT
  - More transferable than simple tight-binding
- Add a self-consistent term for coupling to electrostatic environment
- Vacuum level alignment so that parameter sets can be mixed

Extended Hückel theory (EHT)

Slater type LCAO orbital

$$\phi_{nlm}(\vec{r}) = \frac{r^{n-l-1}}{(2n)!}\left[C_1(2\eta_1)^{2n+1}e^{-\eta_1 r} + C_2(2\eta_2)^{2n+1}e^{-\eta_2 r}\right]Y_{lm}(\hat{r})$$

Overlap matrix

$$S_{ij} = \int \phi_i(\vec{r})\phi_j(\vec{r})dr$$

Hamiltonian

$$H_{ij} = \begin{cases} \varepsilon_i & i = j \\ \frac{1}{2}\beta(\varepsilon_i + \varepsilon_j)S_{ij} & i \neq j \end{cases}$$

M. Wolfsberg, and L. Helmholtz. *J. Chem. Phys.* **20**, 837 (1952)
J. H. Ammeter *et. al.*, *J. Am. Chem. Soc.* **100**, 3686 (1978)
M. H. Whangbo, and R. Hoffmann. *J. Chem. Phys.* **68**, 5498 (1978)

Currently 300 parameters in ATK-SE

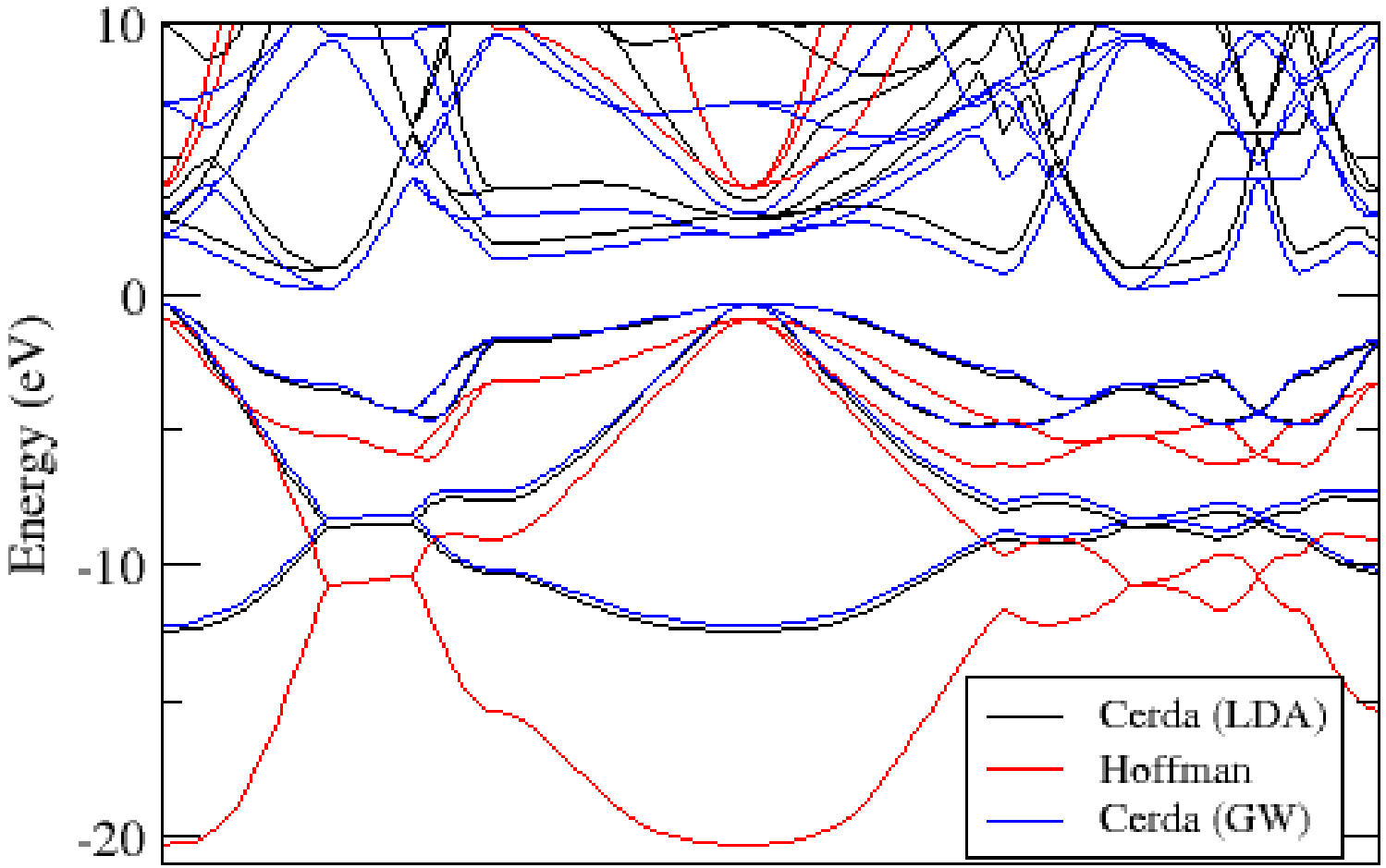
- YAEMOP (Yet Another Extended Huckel Molecular Orbital Program)
  Hoffmann + Muller parameters
  > Fitted to molecules
  > Full periodic system
  > High transferability

M. Wolfsberg, and L. Helmholtz. *J. Chem. Phys.* **20**, 837 (1952)
J. H. Ammeter *et. al.*, *J. Am. Chem. Soc.* **100**, 3686 (1978)
M. H. Whangbo, and R. Hoffmann. *J. Chem. Phys.* **68**, 5498 (1978)

- Jorge Cerda's homepage
  > Special parameters fitted to particular systems

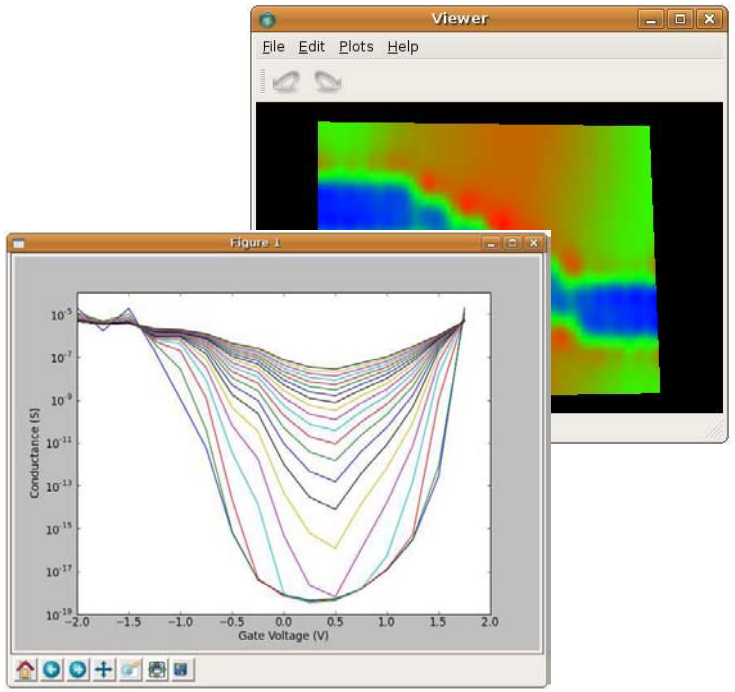
J. Cerda and F. Soria, *Phys. Rev. B* **61**, 7965 (2000)

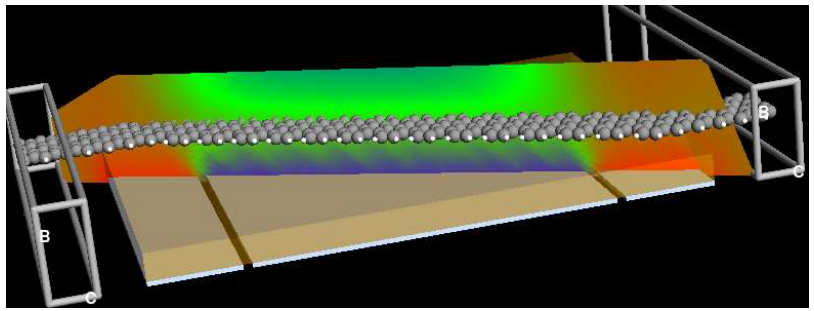
Band structure of silicon, comparison



Electrostatic model

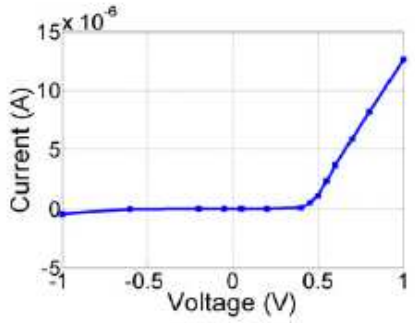
- Support for all Poisson equation boundary conditions in all directions
  - Dirichlet, von Neumann, periodic
  - Multipole for charged (isolated) systems
- Finite bias in transport direction
- Multi-grid (or FFT for periodic)
- Dielectric & metallic regions
  - Arbitrary shape, position, and number
  - Electrodes (metallic; fixed potential/voltage)
  - Screening layers (dielectric constant)
  - No gate currents
  - Described as continua
- Solvent model
  - Ambient dielectric constant
- Non-uniform grids under development

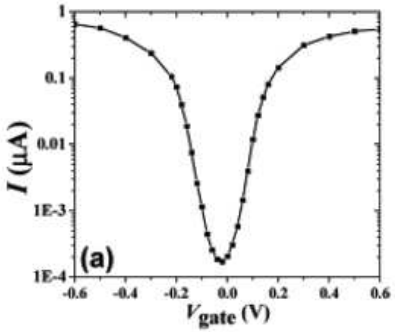


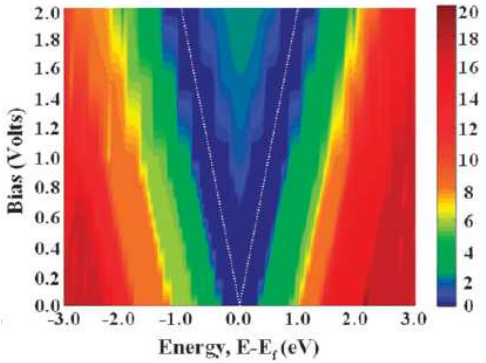


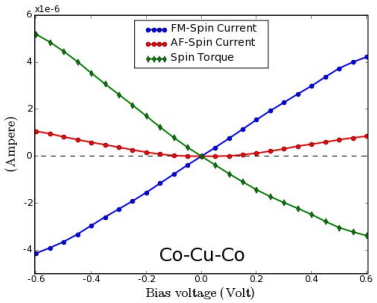
Virtual experiments

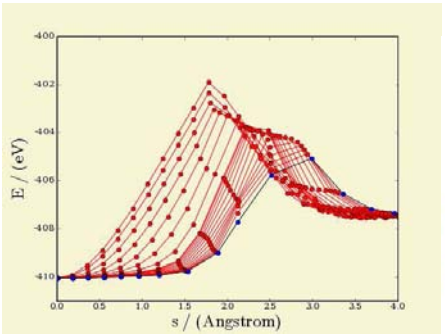
- Current–voltage characteristics
  - NDR, rectification, switching, etc
  - Conductance
  - Tunnel magneto-resistance
  - Spin current
  - Schottky barrier
  - Contact resistance
- Transistor characteristics (gates)
- Voltage drop
- Thermionic emission
- Co-linear spin torque transfer
- Bias-induced forces
- Basic electronic structure
  - Band structure
  - DOS
  - Molecular spectra
  - Total energies
- Reaction pathway & activation energy

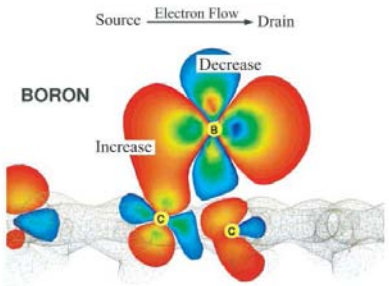






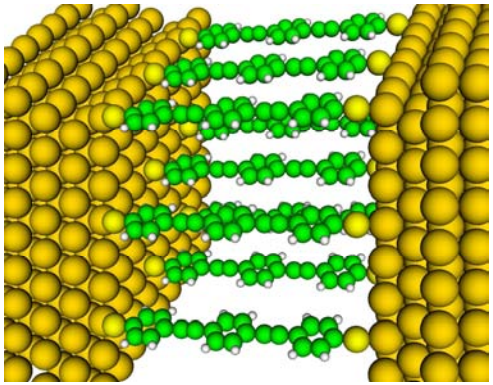


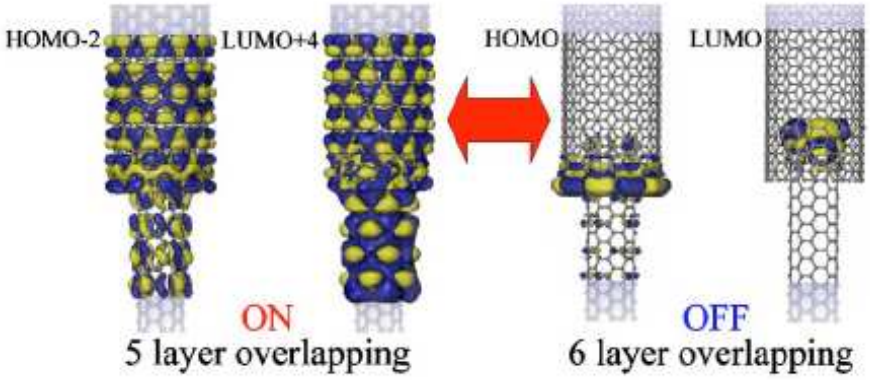


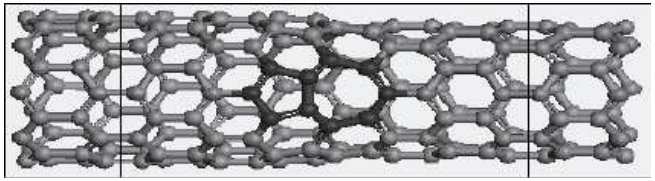


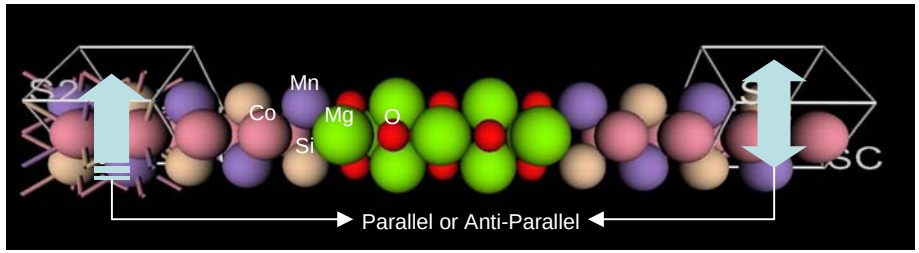
Two-probe systems, electrodes

- A two-probe system consists of
  - left and right electrodes
  - a central scattering region
- Each electrode is a semi-infinite periodic system
  - Cleaved bulk crystal (eg. a gold [111] surface)
  - Carbon nanotube, graphene, atomic chain, etc
- The two electrodes can be different
  - Two different metals
  - Two different carbon nanotubes
  - Metal–nanotube contact
  - Different spin-polarization (magnetic tunnel junctions)





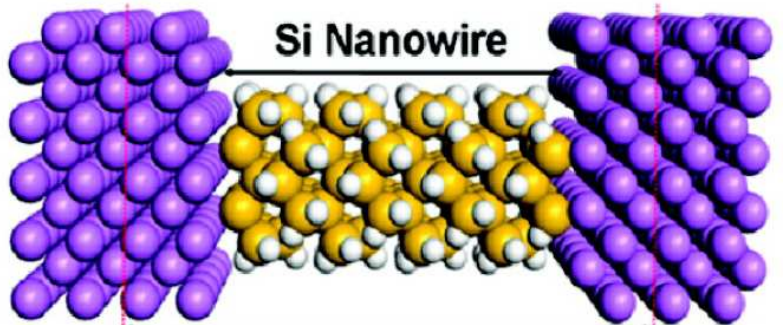


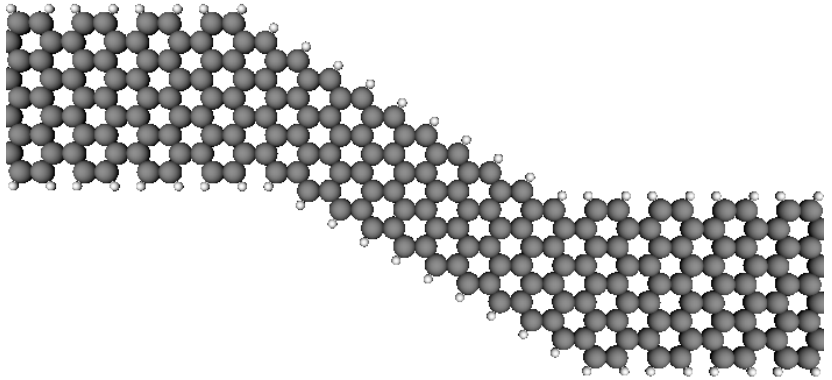


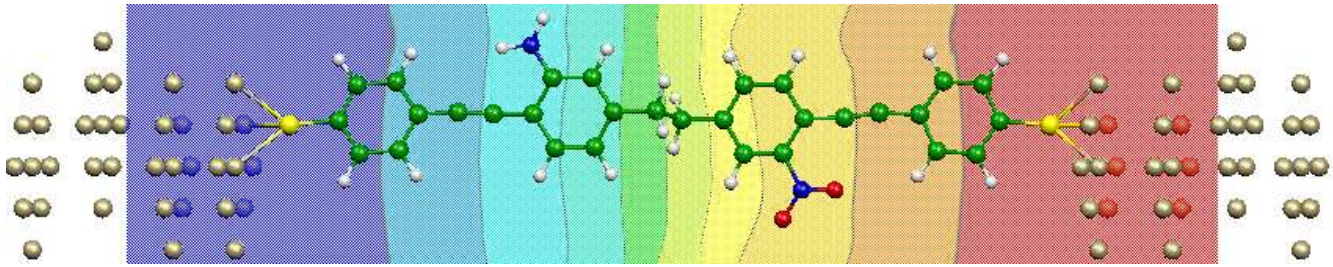
Quantum Wise

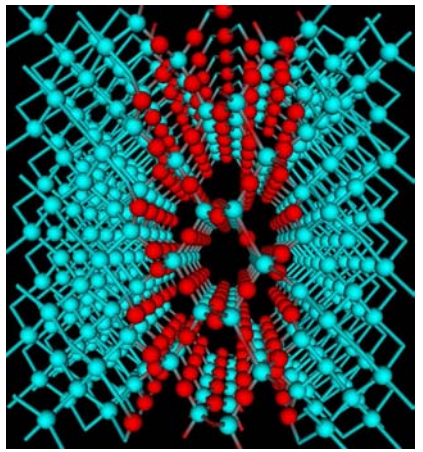
Scattering region

- The scattering region can be ... anything!
  - A molecule (e.g. between two metal surfaces)
  - A piece of a carbon nanotube (e.g. a metal-nanotube-metal contact)
  - A graphene nanoribbon zigzag/armchair contact
  - A periodic structure (layered interfaces)



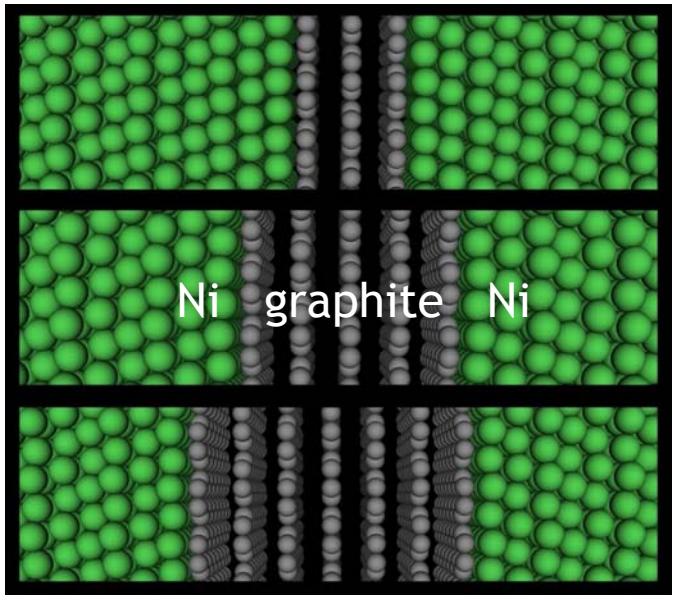


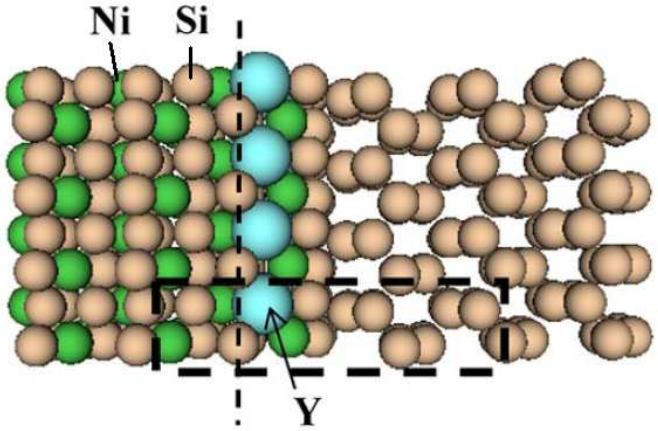


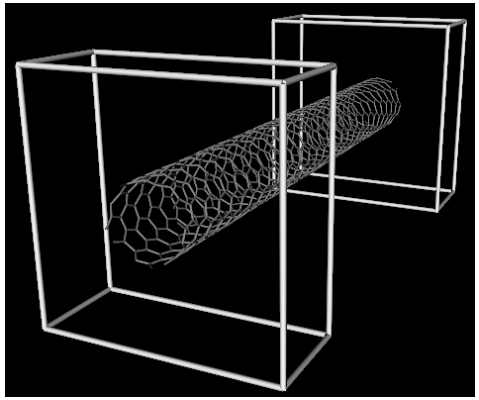


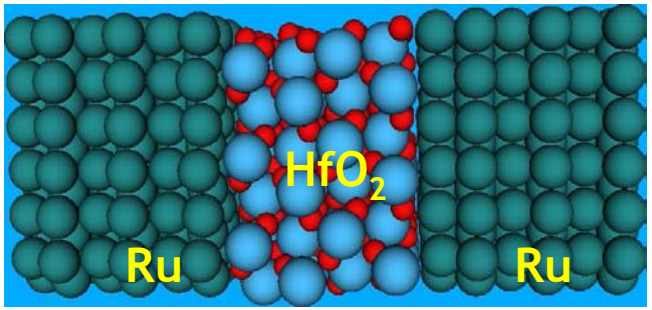
Interface boundary conditions

- Periodic boundary conditions are applied in the transverse plane
  - Makes it simple to model real interfaces, e.g. with surface impurities
  - Vacuum padding for 1D electrodes like nanotubes and wires



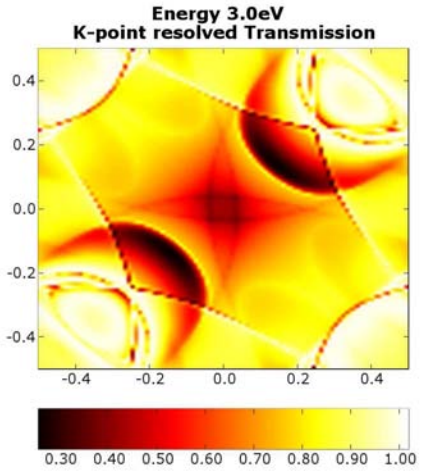


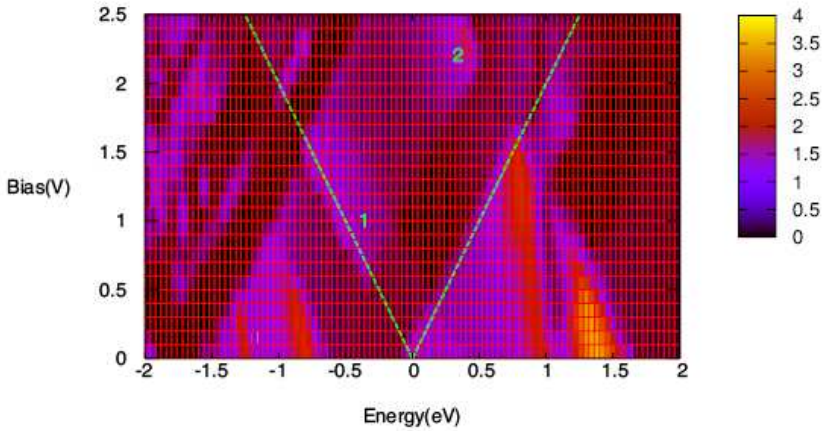


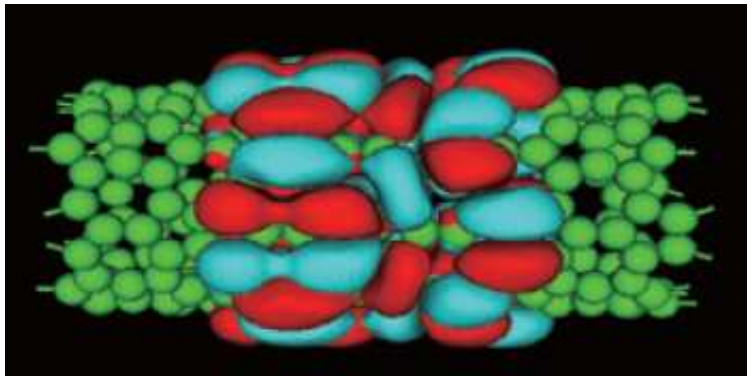


Analyzing quantum transport mechanisms

- ATK provides several analysis tools for understanding transport mechanisms
  - » Transmission spectrum and surface DOS/PDOS
  - » K-point resolved transmission
  - » Molecular projected self-consistent Hamiltonian (MPSH) eigenvalues and eigenstates
  - » Transmission eigenvalues and eigenchannels

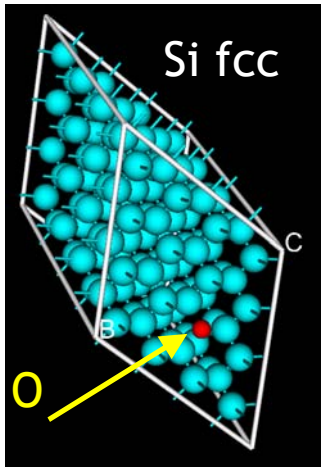


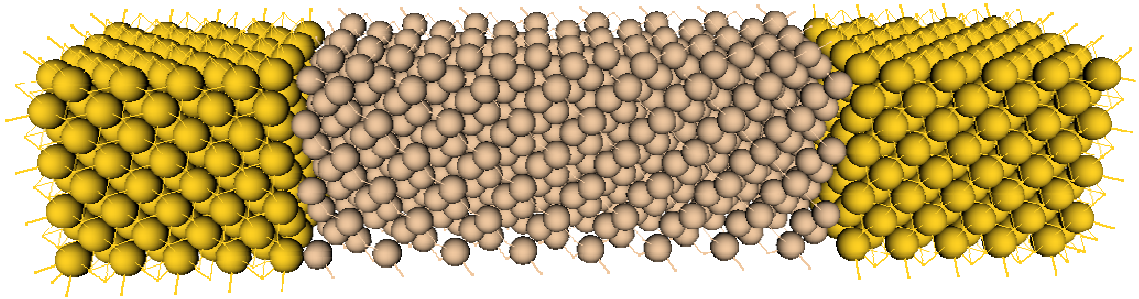


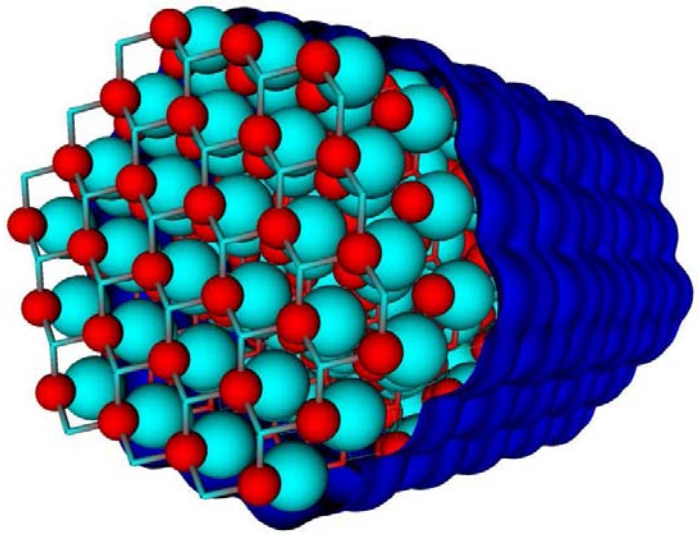


Large-scale systems

- Nanowires, multiwall nanotubes and complex interface structures can easily contain several hundred atoms per unit cell
- In order to treat effects from defects (including doping), very large unit cells are also required
- Need to treat thousands of atoms

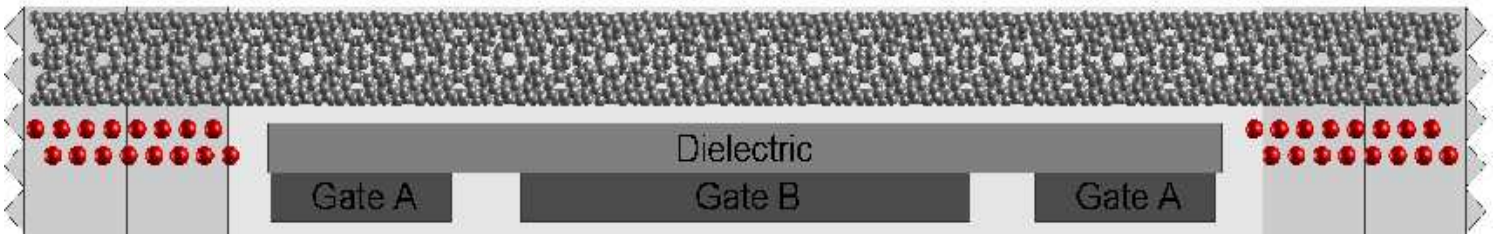


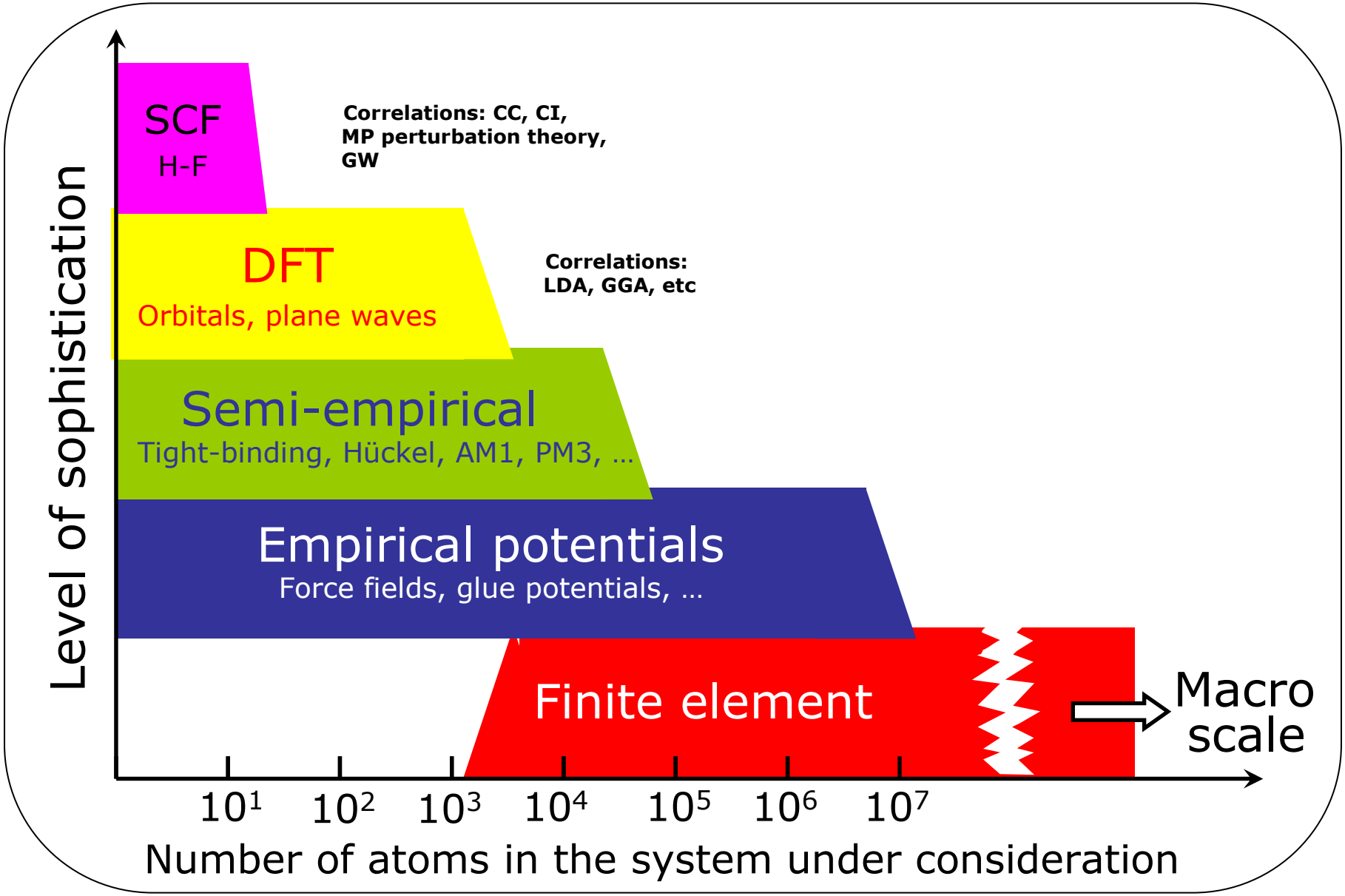




Ways towards larger systems

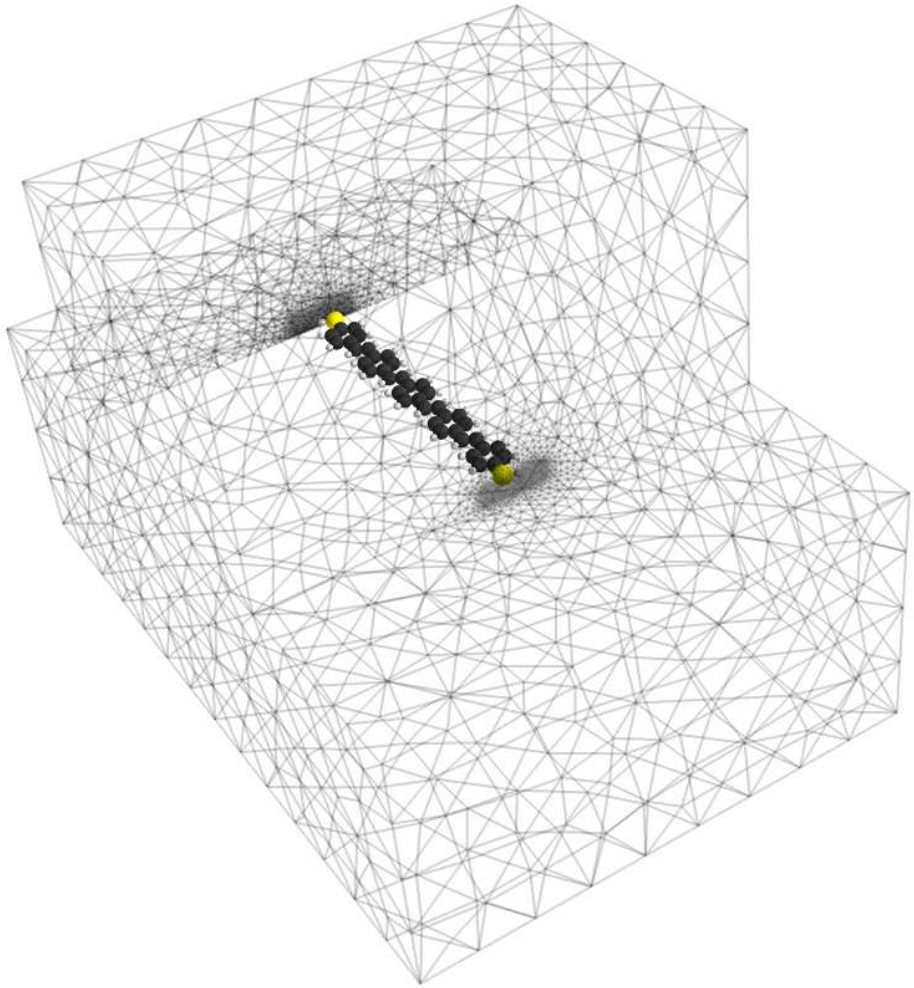
- Method
  - Numerical orbitals allow larger systems than plane waves
  - Less sophistication (doesn't always mean lower accuracy!)
- Multi-scale
  - Coupling to continuum models
  - Extract effective parameters for higher-level models
- Parallelization
  - In particular memory

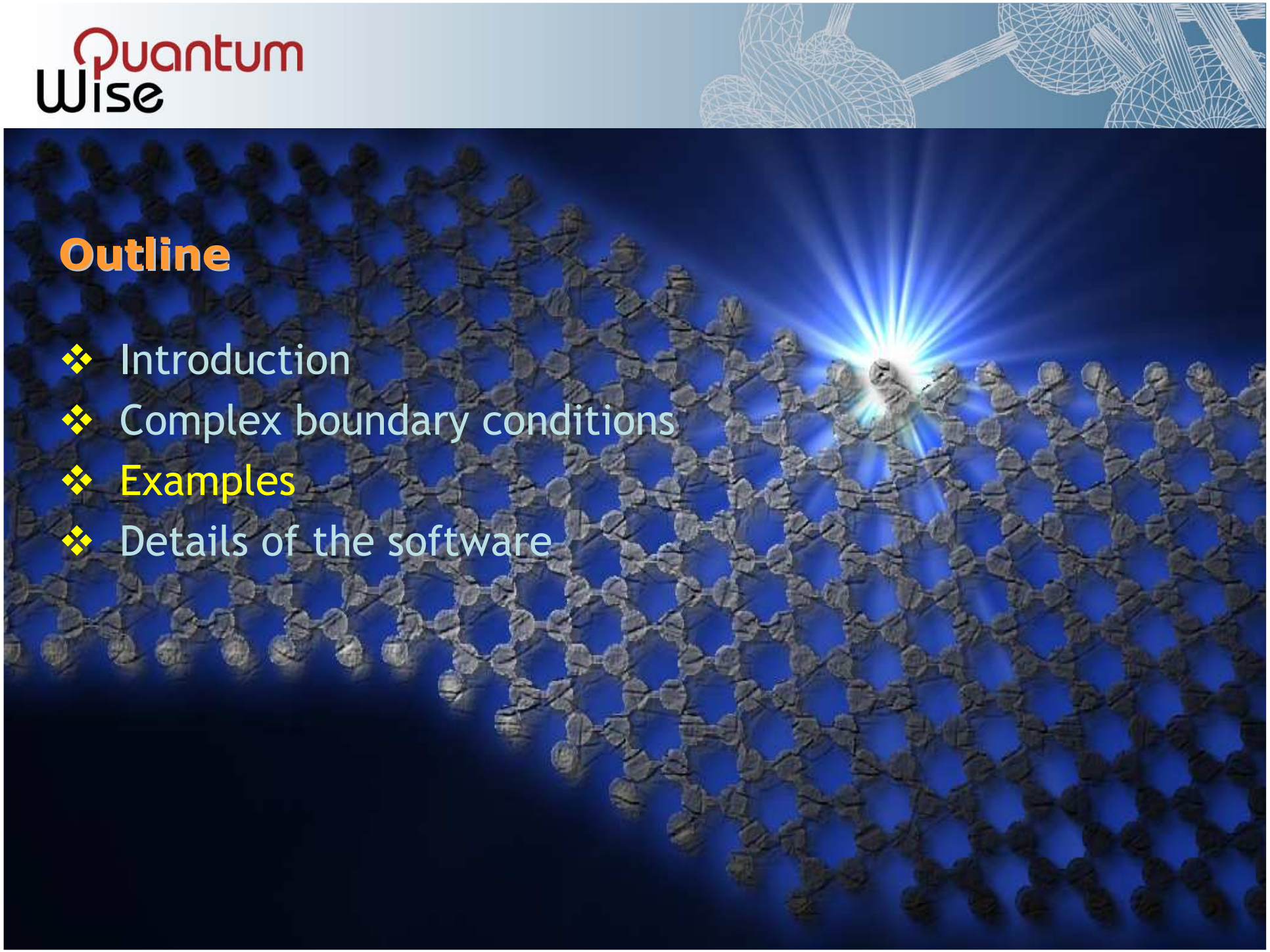



SCF
H-F
Correlations: CC, CI, MP perturbation theory, GW
DFT
Orbitals, plane waves
Correlations: LDA, GGA, etc
Semi-empirical
Tight-binding, Hückel, AM1, PM3, …
Empirical potentials
Force fields, glue potentials, …
Finite element
Macro scale
Level of sophistication
10^1 10^2 10^3 10^4 10^5 10^6 10^7
Number of atoms in the system under consideration

FEM grids

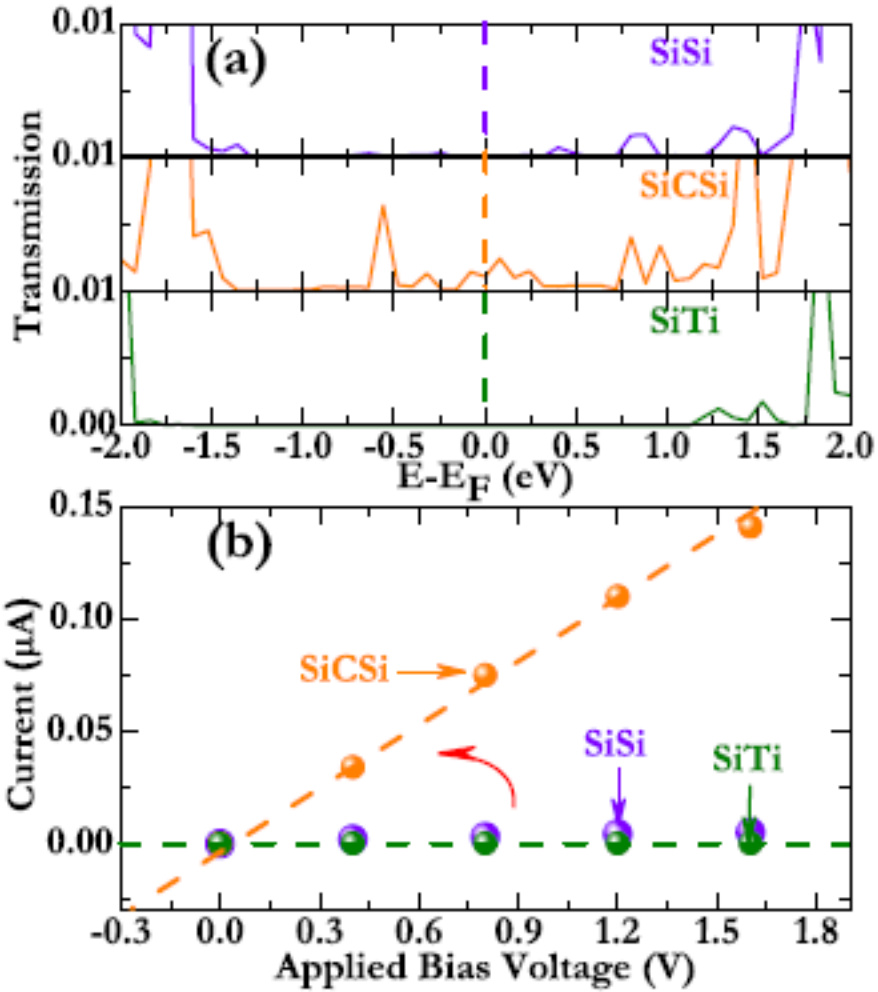
- 3D grids present memory bottle-necks for large systems
- Lots of vacuum for nanotubes & graphene means regular grids are inefficient and wasteful
- Reduce memory & increase performance by using FEM grids
- Point density derived from charge density
- Ongoing project with Copenhagen University

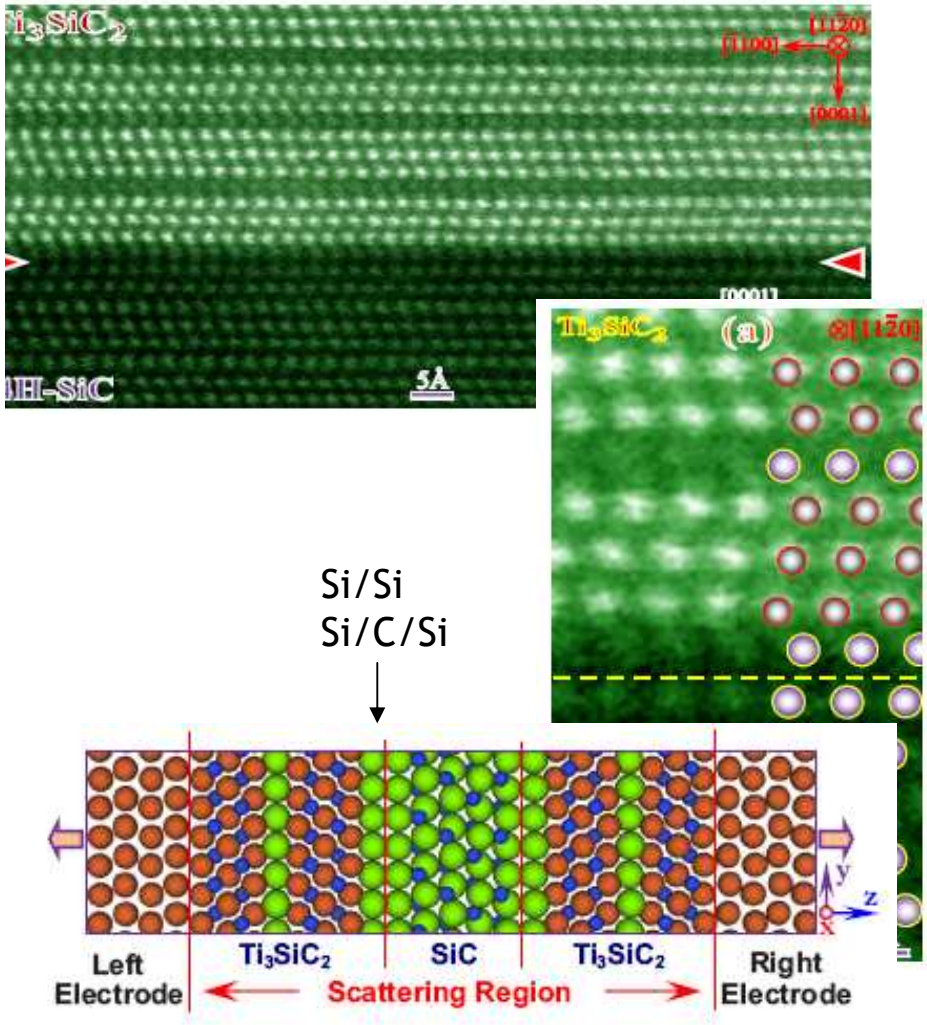



Outline
Introduction
Complex boundary conditions
Examples
Details of the software

Ohmic contacts on silicon-carbide

Wang *et al.*,
PRB **80**, 245303 (2009)
Adv. Mat. **21**, 4966 (2009)

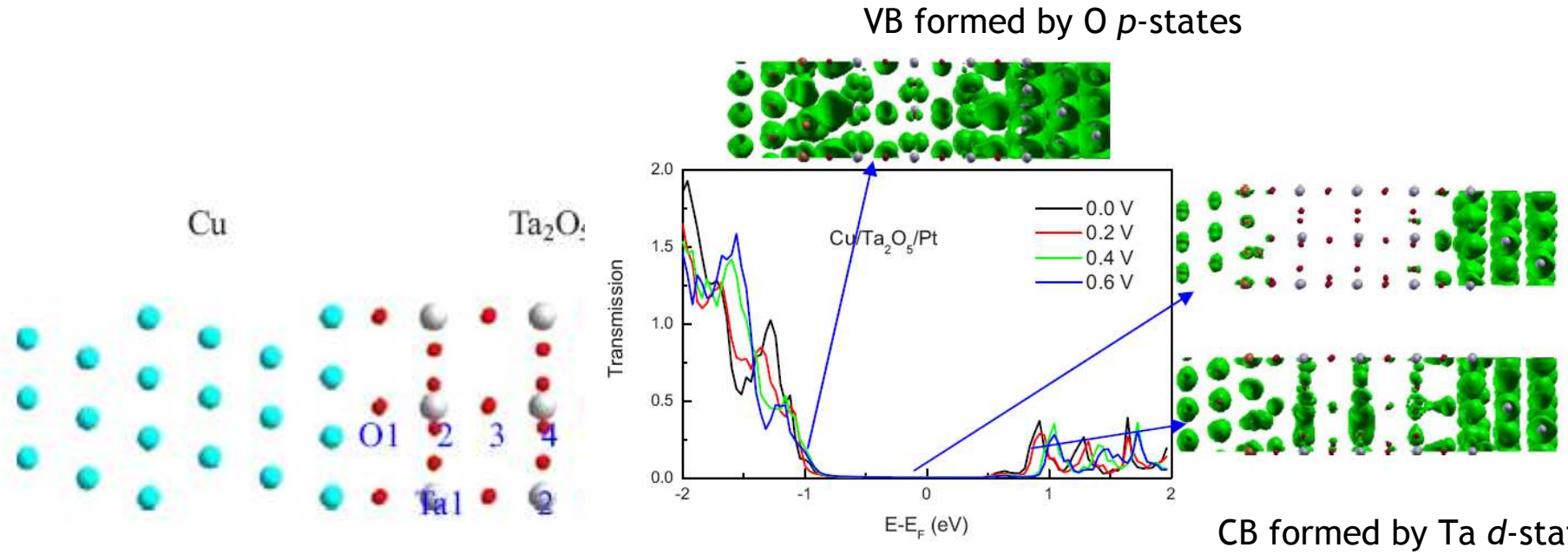




Resistive switch

Gu *et al.*, J. Appl. Phys. **106**, 103713 (2009)

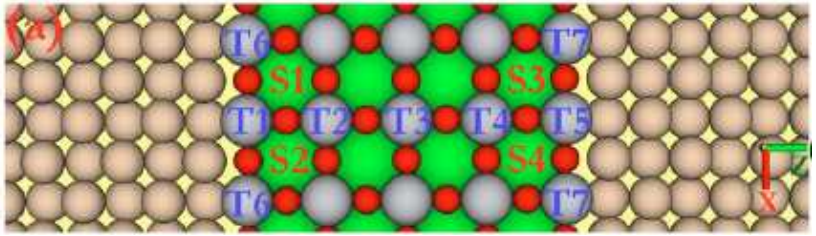
- $Cu/Ta_2O_5/Pt$ heterojunction
- Comparison of β vs δ phase of Ta_2O_5
- No conducting channel formed from Cu to Pt through δ-Ta_2O_5 = “off” state

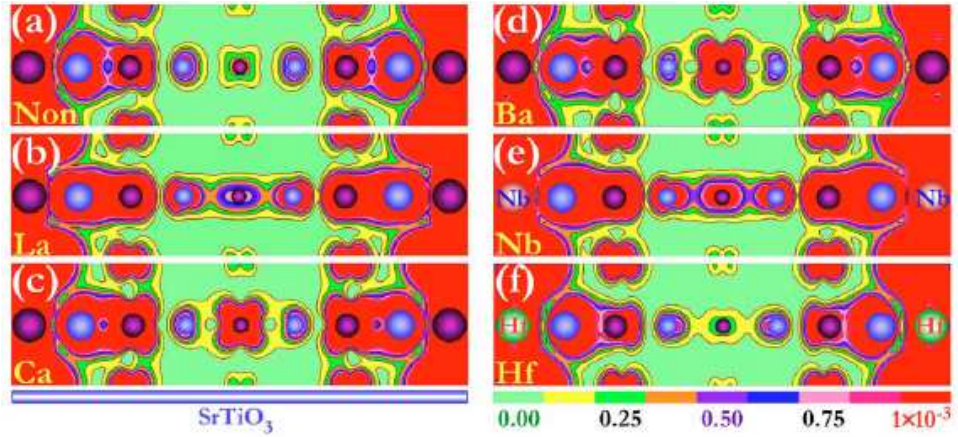


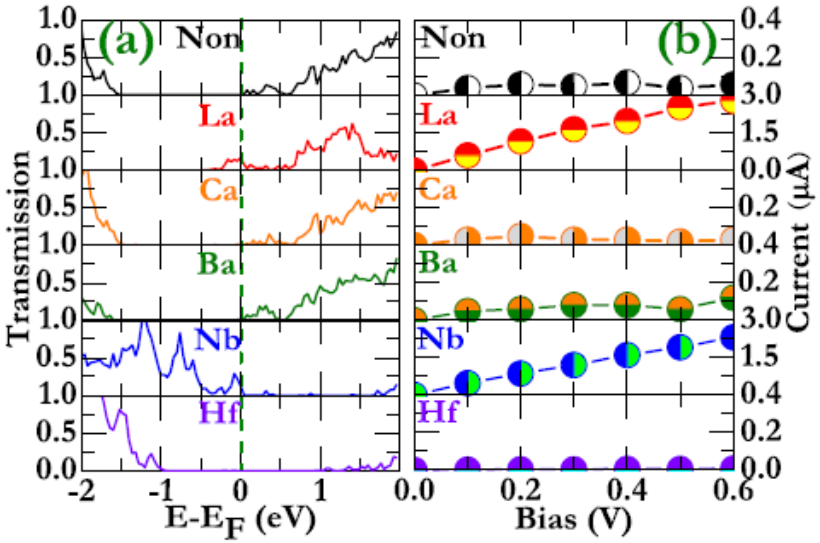
Pt–SrTiO3–Pt heterostructures

Z. Wang *et. al.*, APL **94**, 252103 (2009)

- Intrinsically closed conductance channel in SrTiO3 opens up after doping substitutional atoms of higher valency
- Enhancement in electron transmission at Fermi level, drastic increase in the current with bias







Atomic switch using solid electrolytes

Journal of the Japan Institute of Metals
Vol. 73 (2009) , No. 8 pp.577-582

First Principles Calculations on Electron Conduction Paths in Solid Electrolytes: Toward an Understanding of the Working Mechanism of Atomic Switches

S. Watanabe[1)], T. K. Gu[1)], Z. C. Wang[1)] and T. Tada[1)]

1) Department of Materials Engineering, Graduate School of Engineering, The University of Tokyo

(Received May 1, 2009)
(Accepted June 1, 2009)

As a first step to clarify the microscopic working mechanism of atomic switches using solid electrolytes, we have examined electronic states and electron transport properties of Ag-Ag_2S-Ag and Cu-Ta_2O_5-Pt systems using standard density functional theory and novel non-equilibrium Green's function method. In both the systems, we found that the bridge structures consisting of excess Ag or Cu in the middle solid electrolyte layer work as electronic conduction paths. In Cu-Ta_2O_5-Pt, we also found that O vacancy is much less effective for the enhancement of electronic conduction than excess Cu.

Silver sulfide atomic switch

Wang *et al.*, APL **93**, 152106 (2008)

- Ag_2S atomic switch from an $Ag|Ag_2S|Ag$ heterojunction
- Conductive, metallic bridge formed from excess silver in Ag_2S

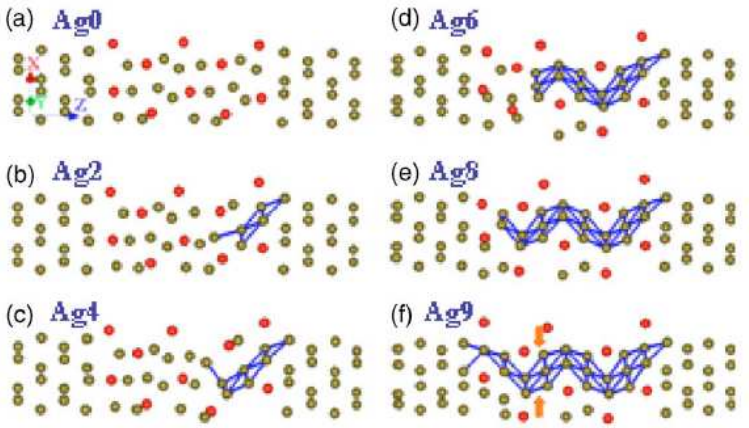
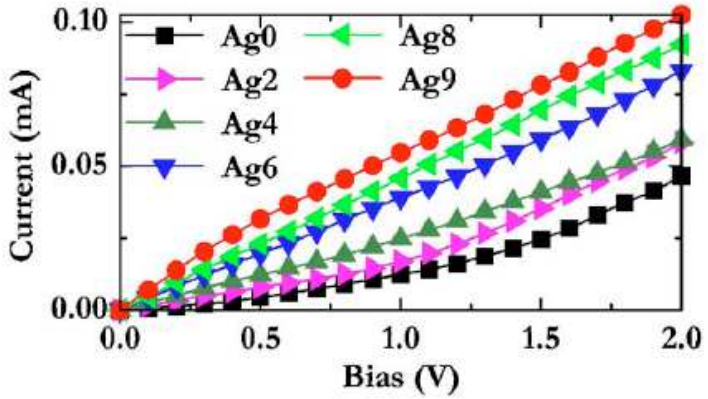


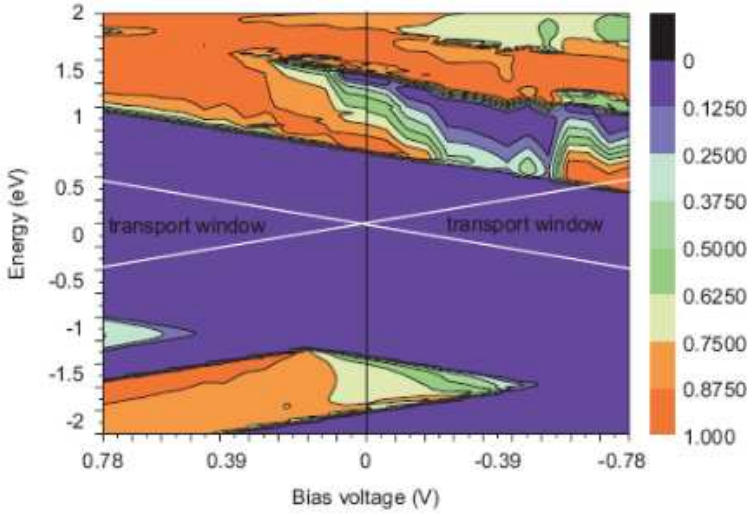
FIG. 2. (Color online) Evolution of a Ag bridge formation inside the Ag_2S with the insertion of excess Ag atoms. Six representative configurations, (a)–(f), are presented. The formed bridge is linked by lines.

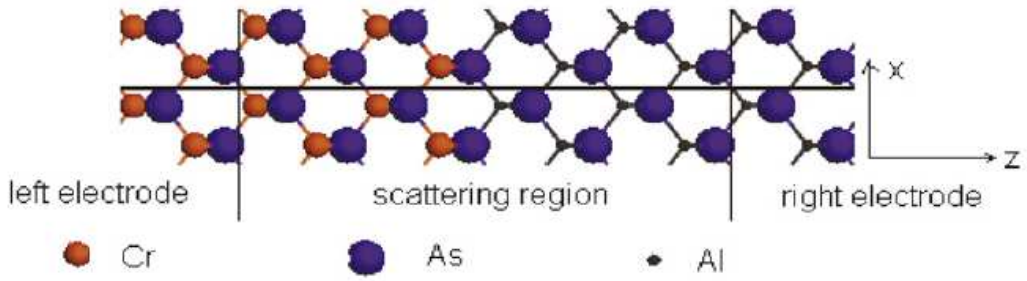


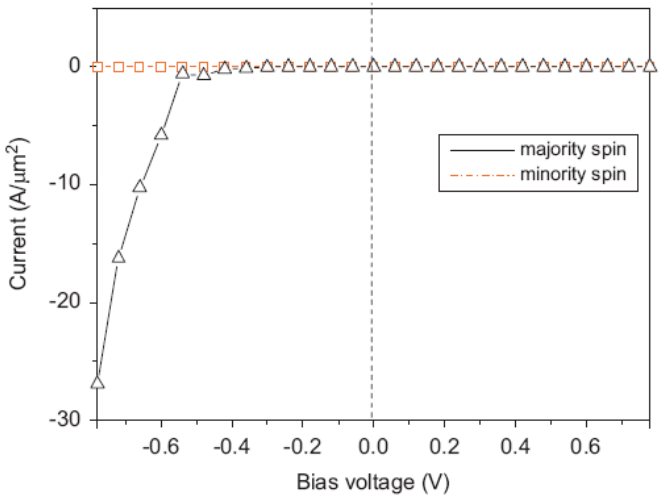
Spin-dependent transport

Y. Min *et al.*, Journal of Magnetism and Magnetic Materials **321**, 312 (2009)

- CrAs(001)/AlAs(001) heterogeneous junction
- Strong diode effect of charge and spin current

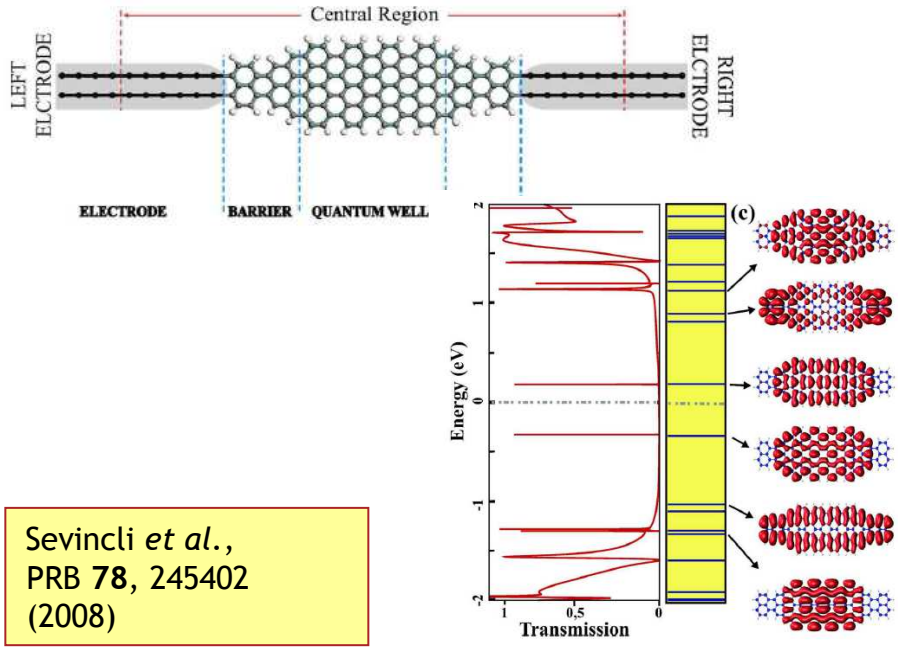






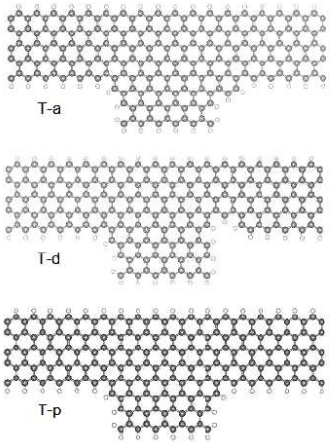
Shapes of graphene

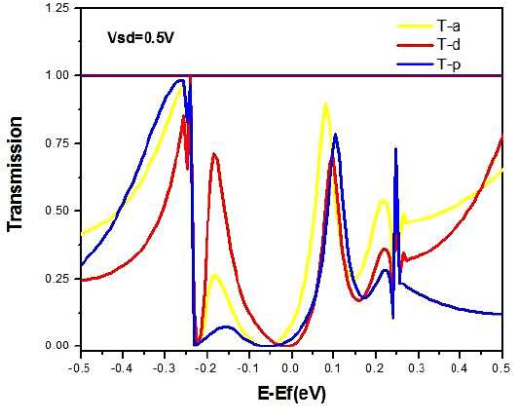
- Resonant-tunneling double-barrier diode (RTDB)
- Commensurate graphene or graphene/B-N armchair structures



Sevincli *et al.*, PRB **78**, 245402 (2008)

- T-shaped junctions
- Current depends strongly on stem height
- Conductivity controllable by selective doping

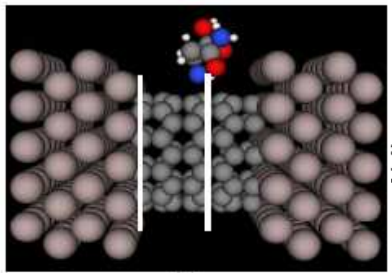


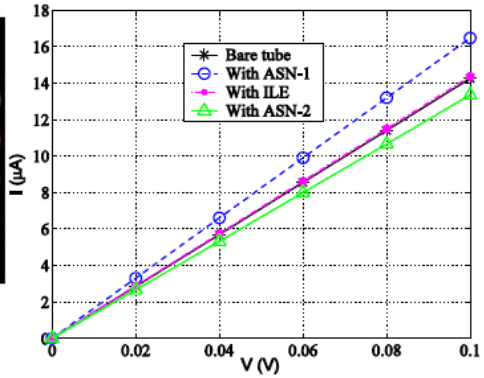


OuYang *et al.*, Nanotechnology **20**, 055202 (2009)

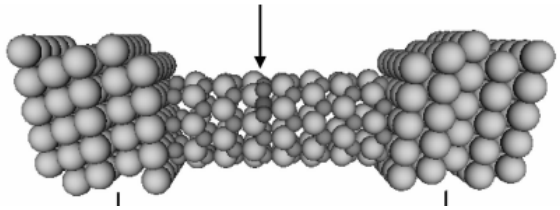
Gas sensing

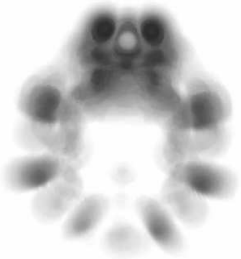
- Amino acids by CNTs





- NO_2 by SiC nanotubes



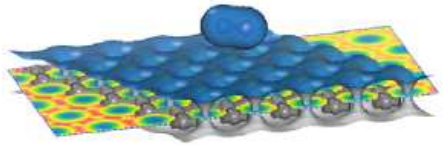


Abadir *et al.*,
Int. J. High Speed Electr. Syst., **18**, 879 (2008)
8th IEEE Conf. Nanotech., 230 (2008)

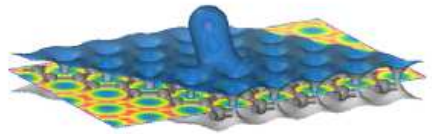
Ruixhue *et al.*, J. Semicond. **30**, 114010 (2009)

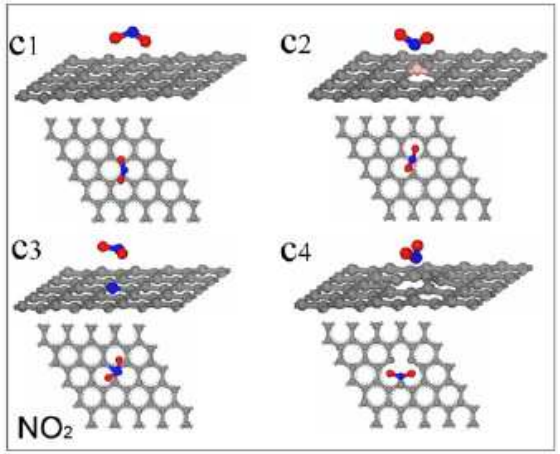
- By graphene (CO, NO, NO_2, NH_3)
- Defect/doped vs. pristine

(a) CO on P-graphene



(b) CO on D-graphene

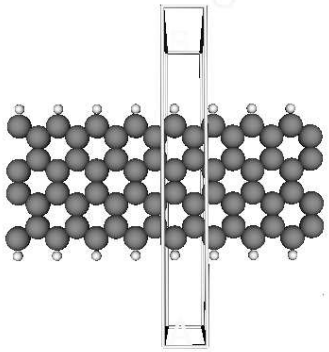


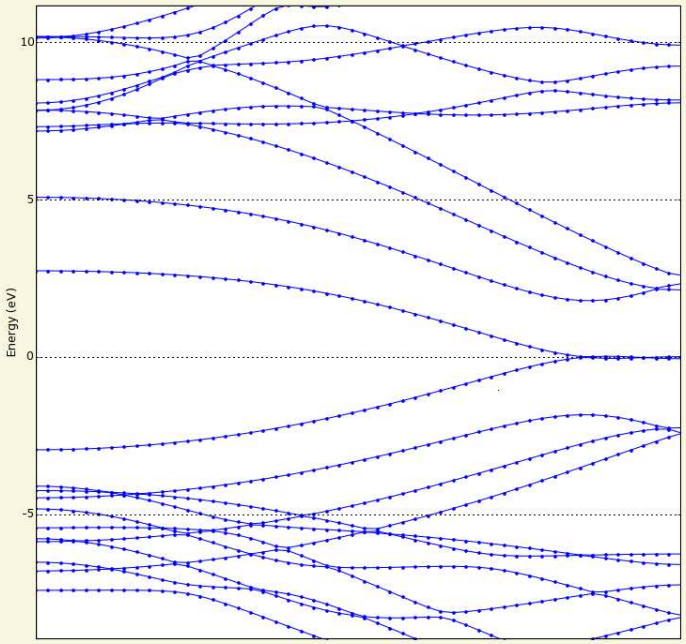


Zhang *et al.*,
Nanotechnology
20, 185504 (2009)

Spin in graphene nanoribbons

- Zigzag nanoribbons are metallic if spin is neglected





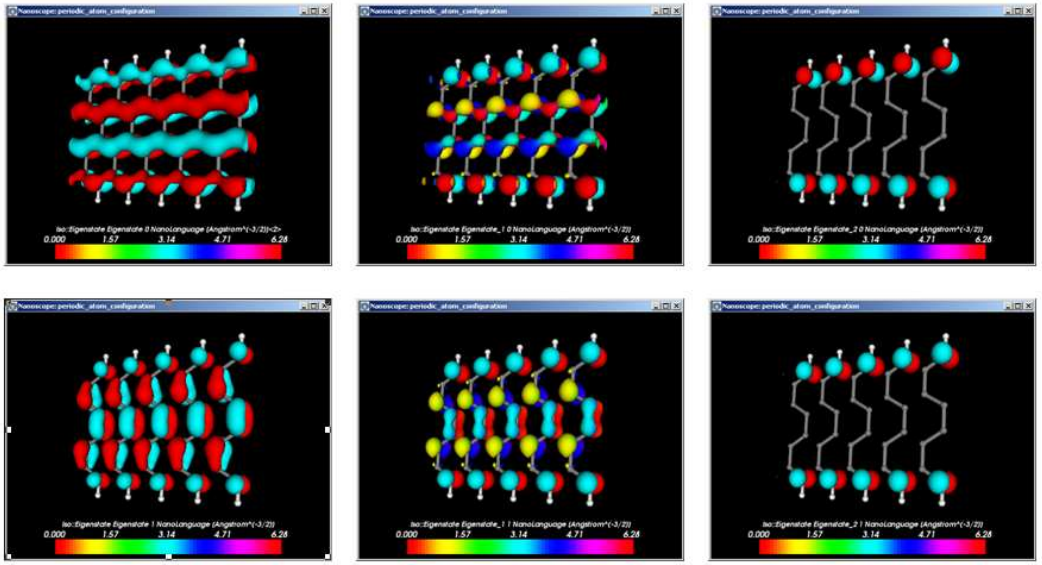


Figure 8: From left to right, the Γ point, $k = 0.35$, and the Z point. The top row are the valence band states, the bottom row the conduction band.

Spintronics in graphene

- Anti-ferromagnetic spin configuration
- Spin opens up a band gap
- Spin states localize on opposite sides of the ribbon
- Majority/minority still fully degenerate

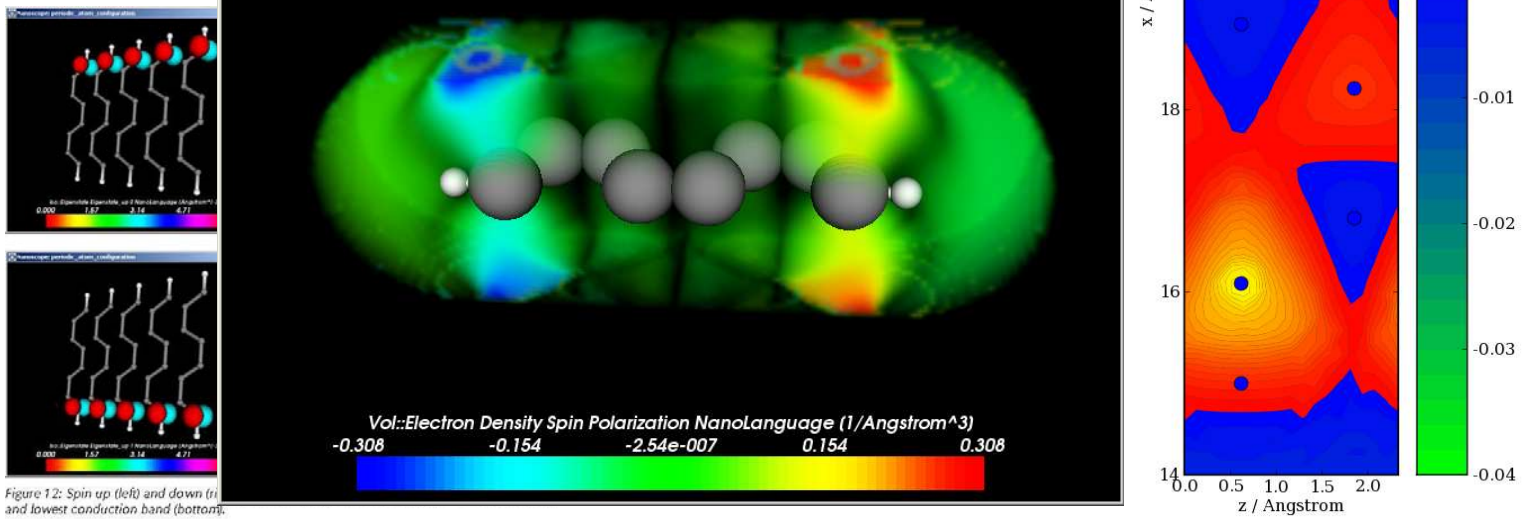
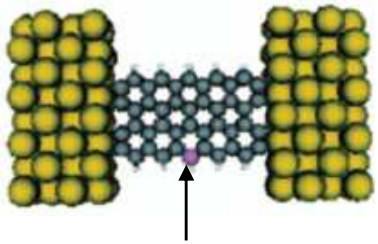
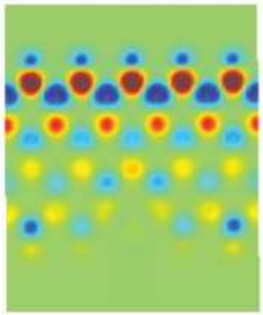


Figure 12: Spin up (left) and down (r
and lowest conduction band (bottom).

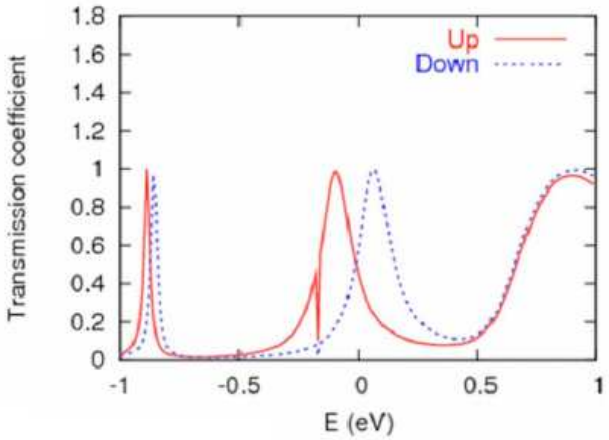
Spintronics with graphene nanoribbons

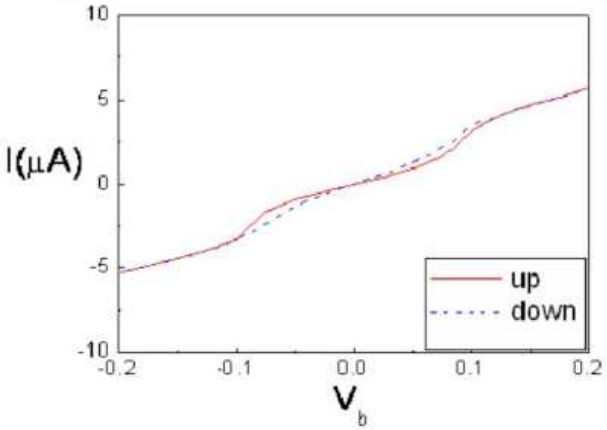
- Graphene between gold surfaces
- Investigation of both the FM and AM states
- Non-magnetic impurities (boron)
- Influence of edge vs. center doping on magnetization, band gap, and spin transport





Edge-doped (B) GNR (FM)



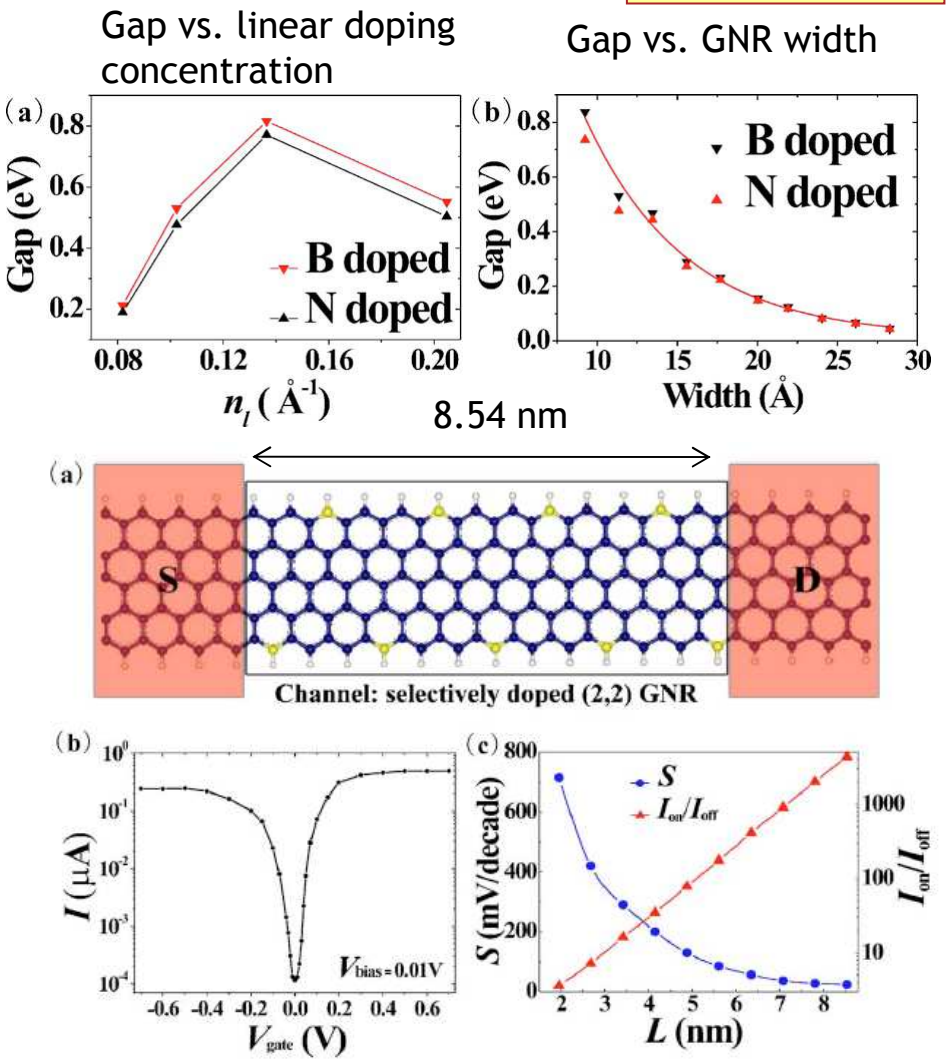


Park *et al.*, J. Chem. Phys. **130**, 214103 (2009)

GNR field effect transistor (FET)

B. Huang *et al.*, APL **91**, 253122 (2007)

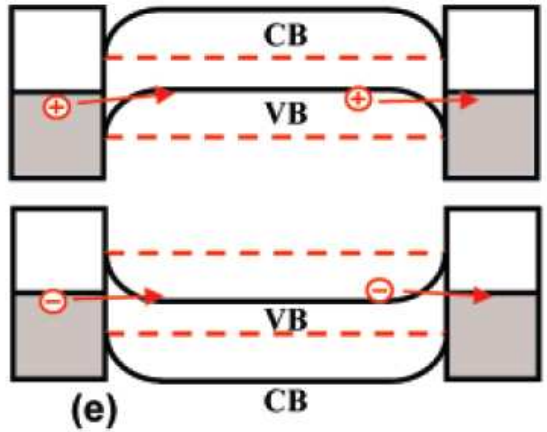
- Challenge to fabricate GNR with energy gap > 0.2 eV (to reduce leakage current)
- Substitutional edge doping (B,N) → band gap 0.8 eV
- N-doped FET exhibits ambipolar characteristics with on-current 1 μA.
- Minimum leakage current is limited to $1.2*10^{-4}$ μA
- High on/off current ratio *~2000*
- Excellent "theoretical" switching characteristics: sub-threshold swing ~40 mV/decade

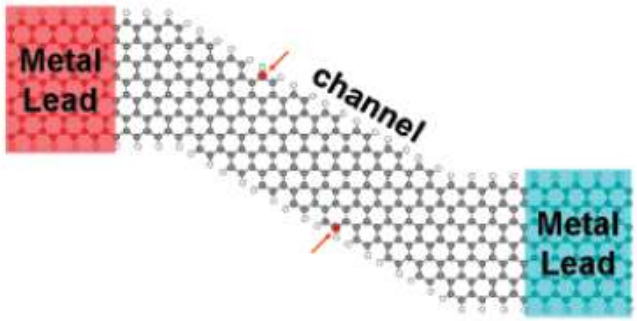


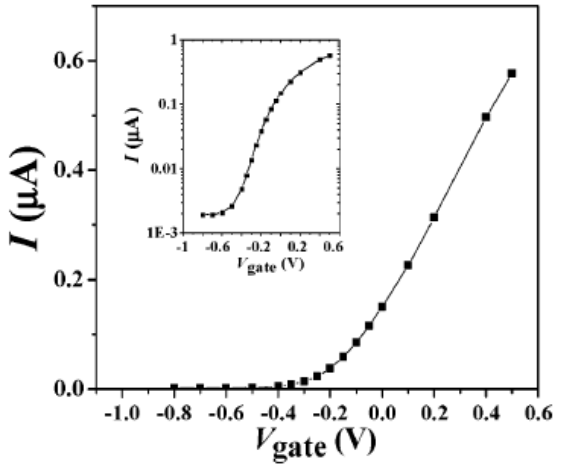
Z-shaped GNR FET

Q. Yan *et al.*, NanoLetters **7**, 1469 (2007)

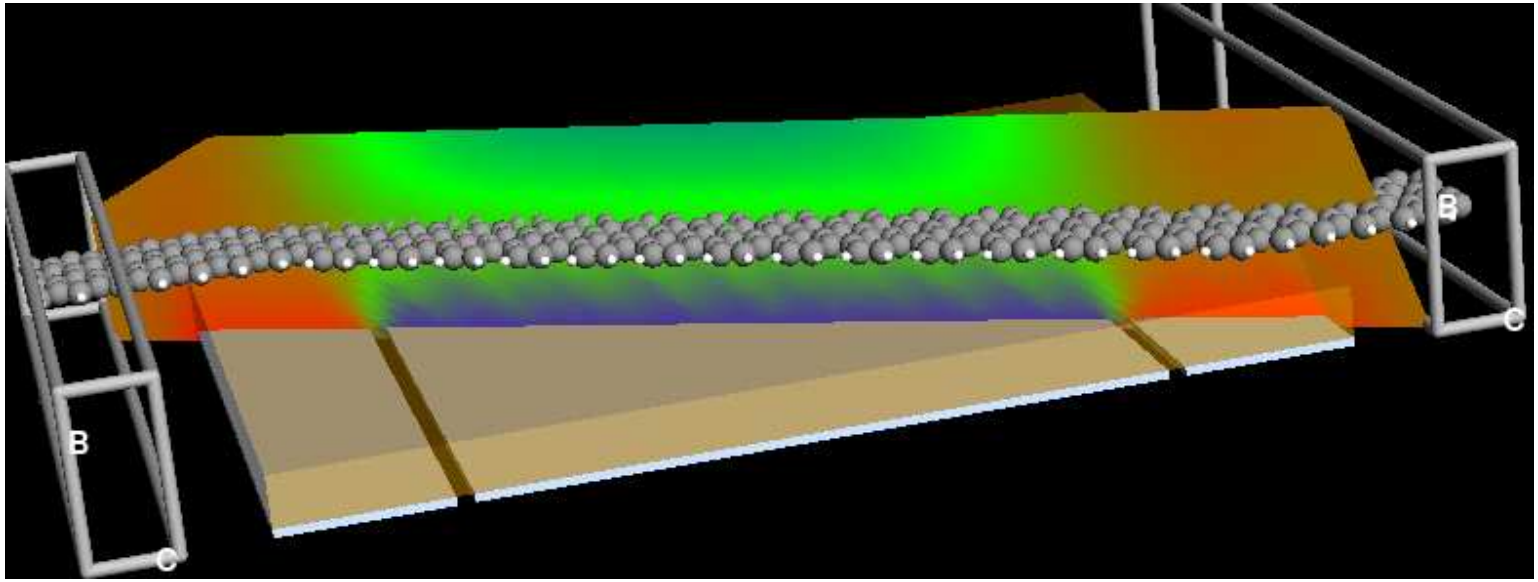
- Armchair segment between zigzag electrodes creates a metal-semiconductor-metal junction
- First-principles modeling allows studies of the influence of edge doping concentration
- Influence of gate voltage
- FET characteristics:
  - ON/OFF ratio $10^{3\text{-}4}$
  - subthreshold swing 60 meV per decade
  - transconductance $9.5*10^3$ S/m.







Graphene junction device – a nanotransistor



Detailed tutorial

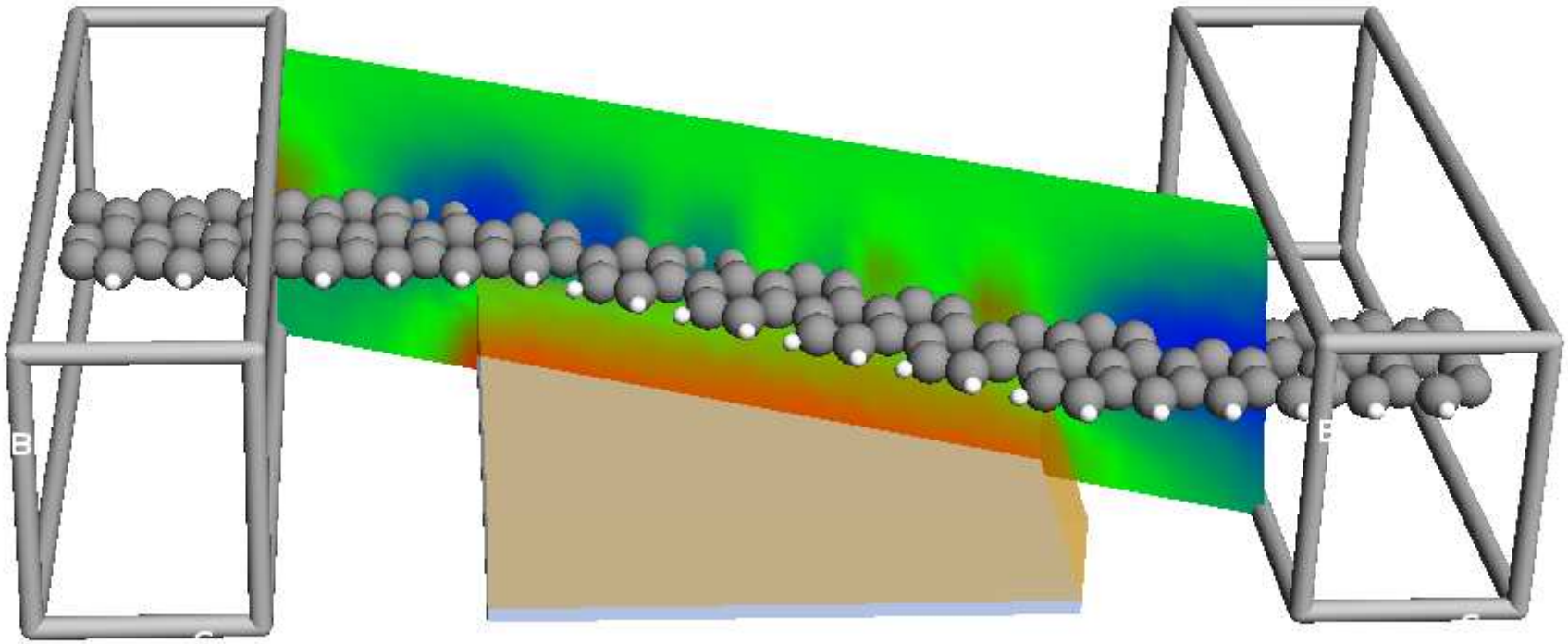
- Current as function of source–drain bias
- Influence of gate bias on conductance
- Influence of temperature on conductance
- Comparison of linear response vs. fully self-consistent calculation

Inspired by Q. Yan *et al.*, Nano Letters **7**, 1469 (2007)

Dependence on the gate potential

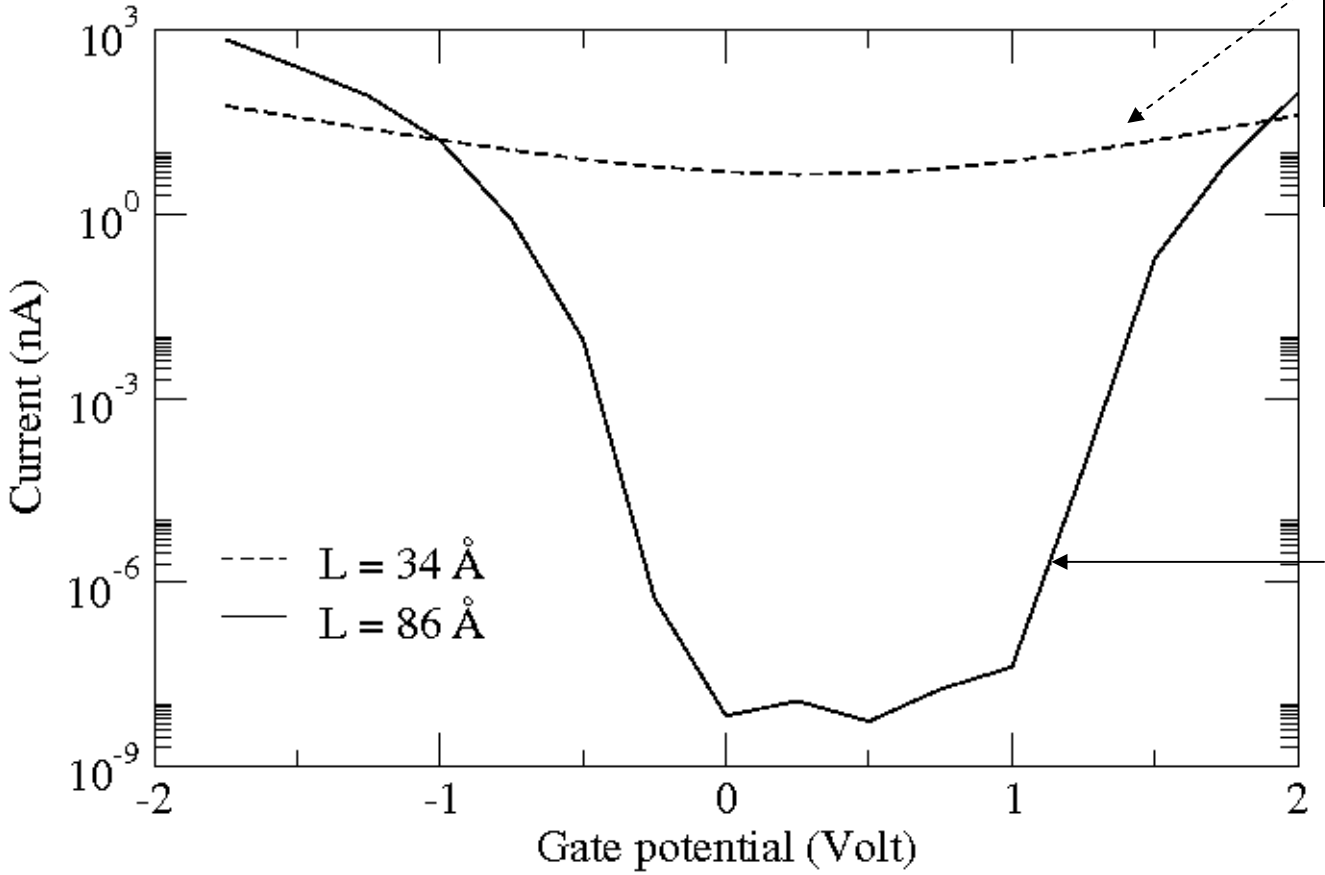
V_{source}=-0.25 Volt

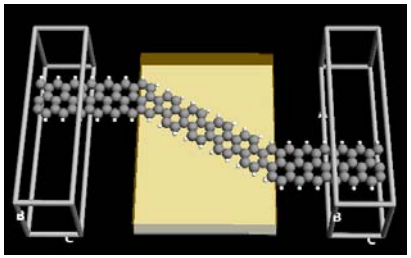
V_{drain}= 0.25 Volt



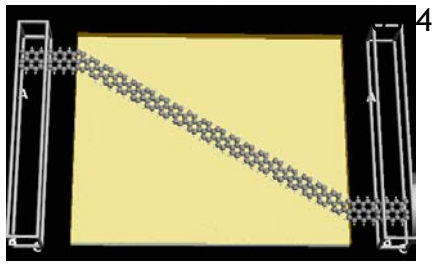
V_{gate}=- 1.0 Volt

Dependence of current on gate potential



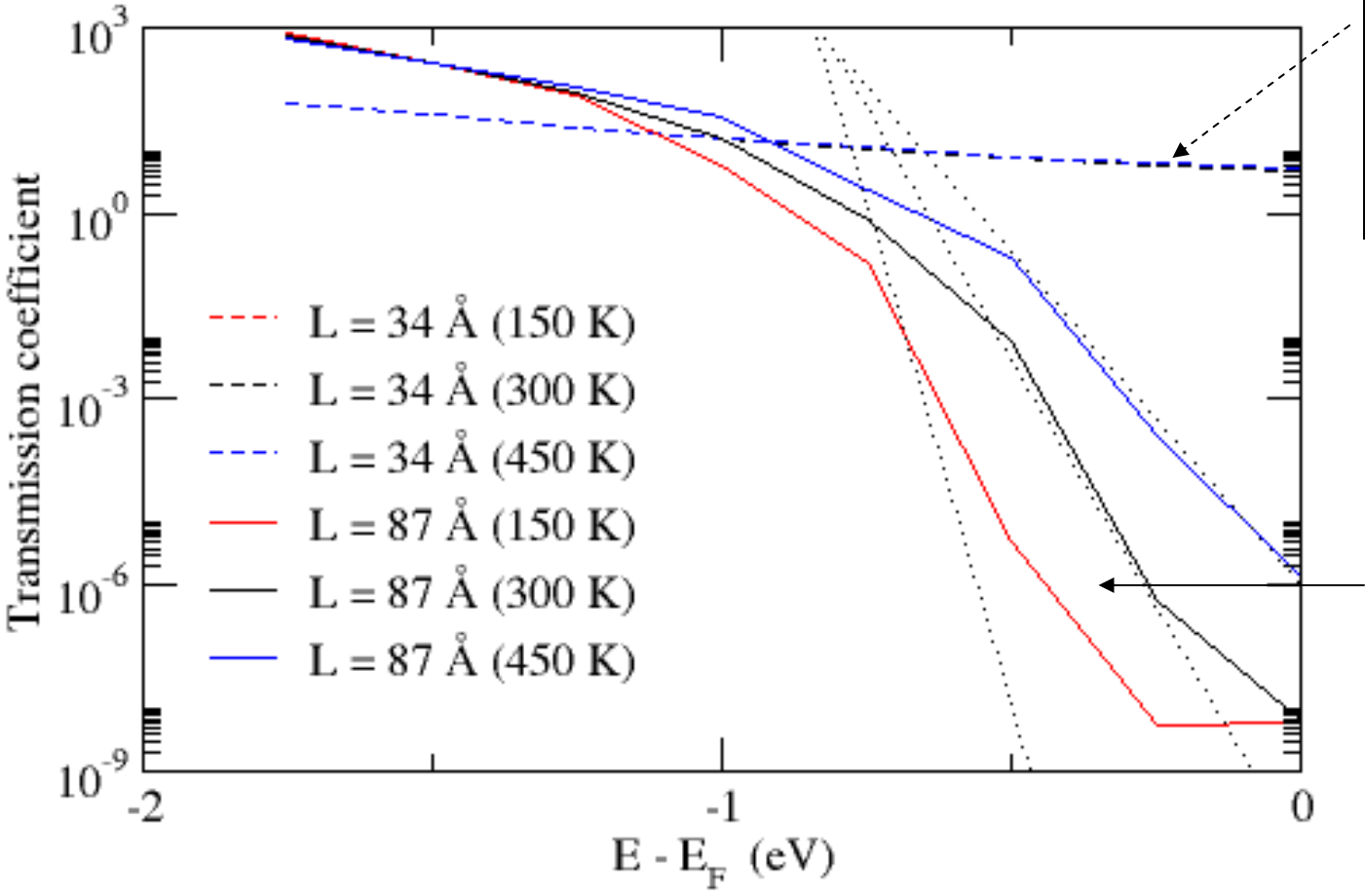


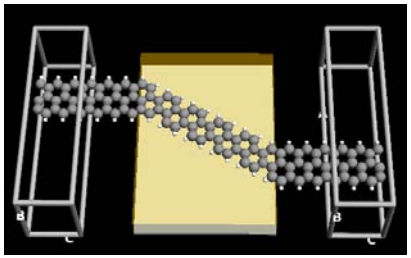
ON/OFF Ratio ~ 10



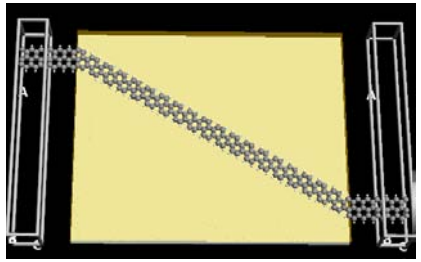
ON/OFF Ratio ~ 10^{9}

Dependence of transistor characteristics on temperature





No temperature dependence



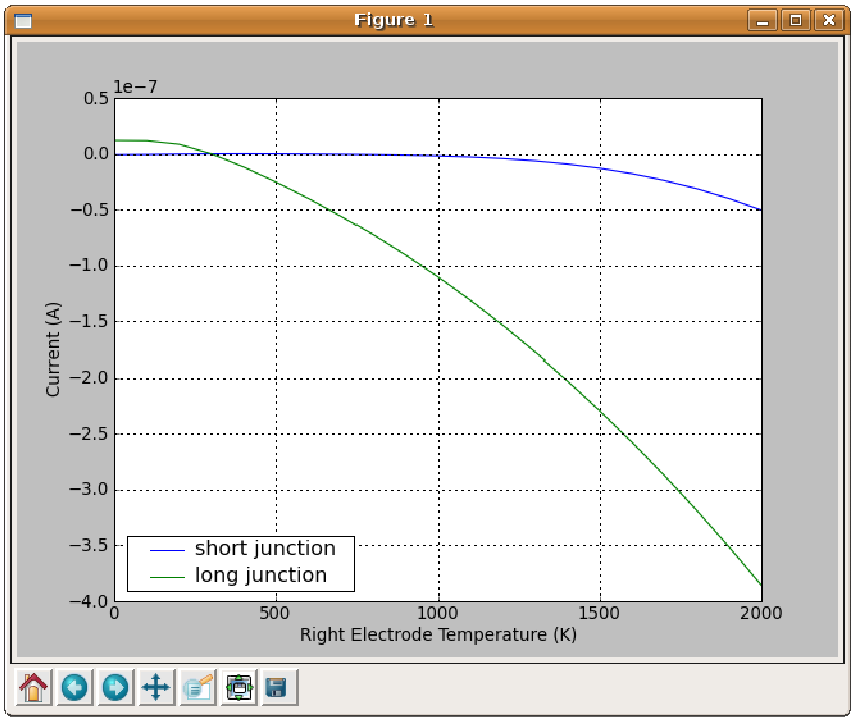
Gate characteristics follows $1/k_bT$

Temperature-dependent conductance

- Short junction: pure tunneling
- Longer junctions: thermionic emission
- Termoelectric effect (current at zero bias)

$$\sigma(T_L) = \frac{e^2}{h}\int T(E) f'\left(\frac{E-(E_F^L)}{k_b T_R}\right)\frac{dE}{k_b T_R}$$

$$I = \frac{2e}{h}\int_{-\infty}^{\infty} d\varepsilon T(\varepsilon)[n_F(\varepsilon-\mu_L) - n_F(\varepsilon-\mu_R)]$$

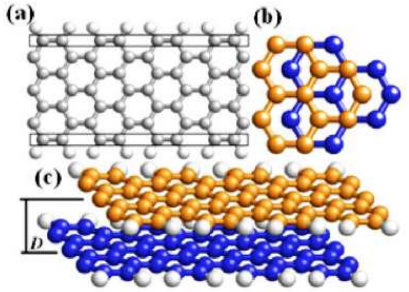


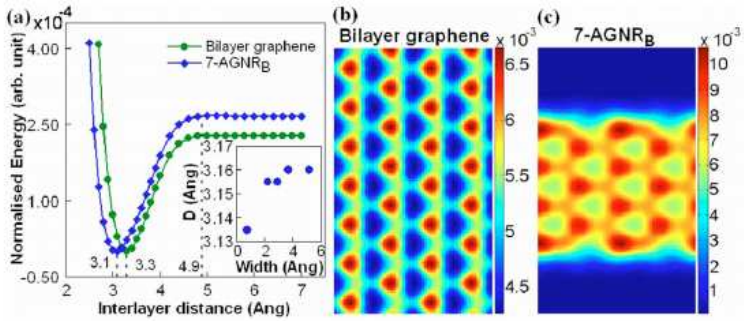
Bilayer structures

Lam *et al.*, APL **95**, 143107 (2009)

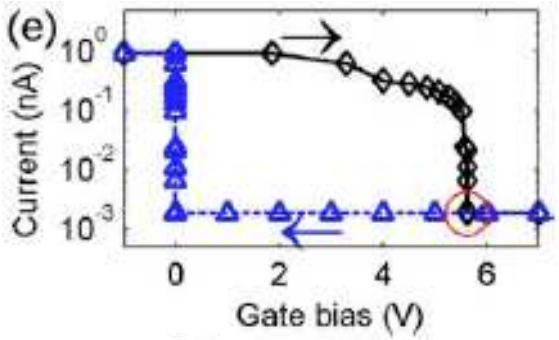
- Armchair bilayer ribbon
- Gap vs. interlayer separation
- Edge doping

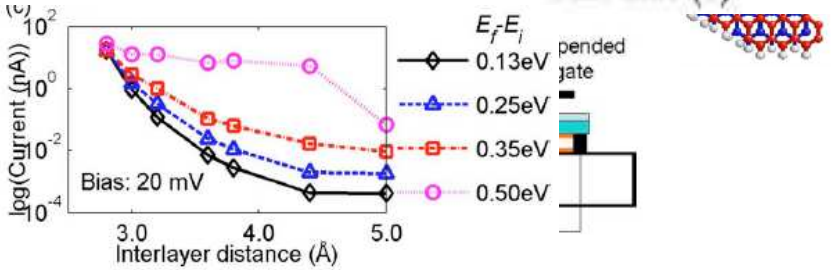
Lam & Liang, APL **92**, 223106 (2008)



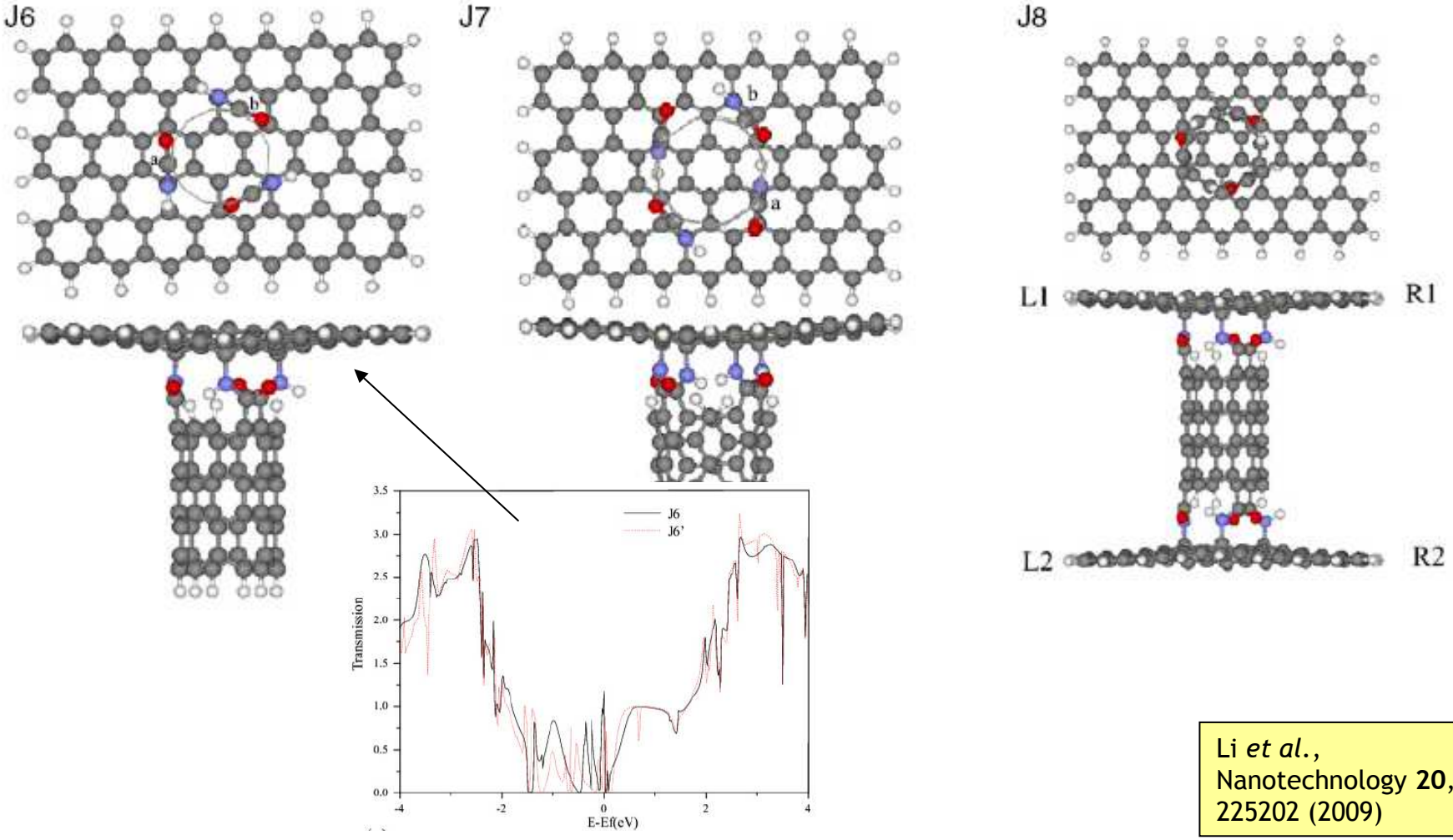


- Nanoelectromechanical force/displacement sensor
- On/off:10^3 @ 20 mV bias and 5.6 V switching gate bias



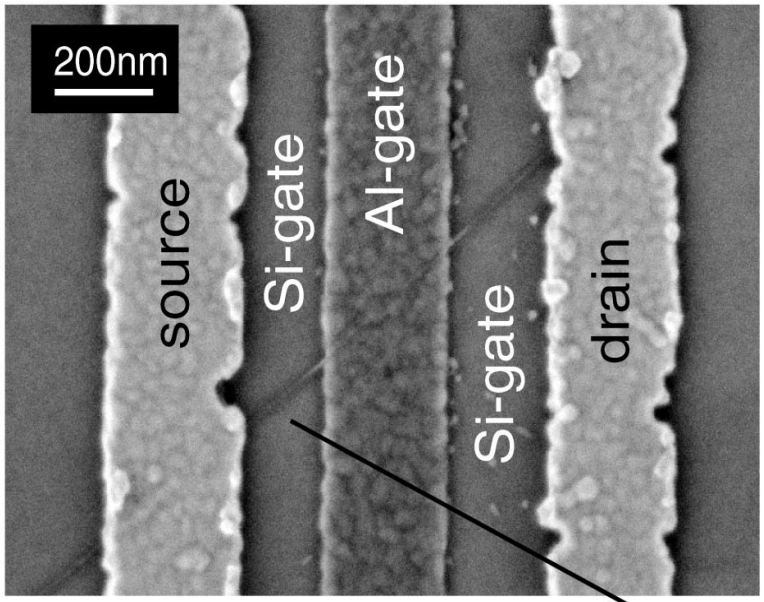


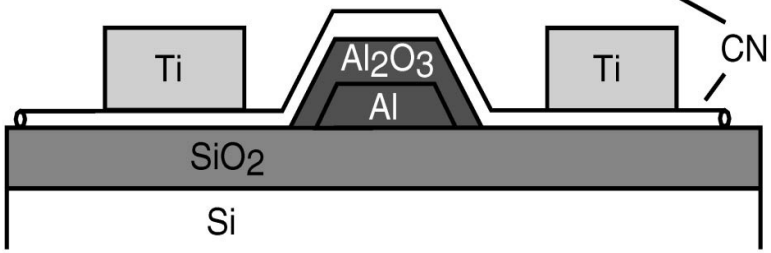
CNT/GNR junctions



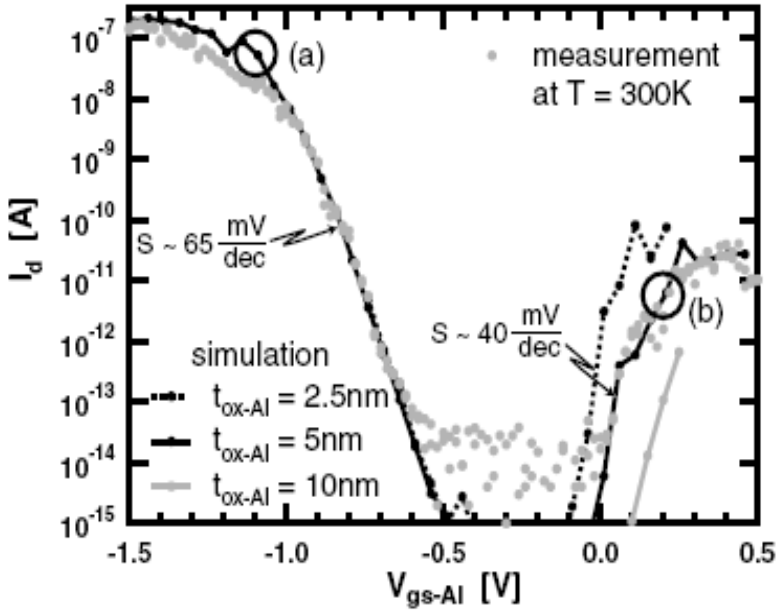
Li *et al.*, Nanotechnology **20**, 225202 (2009)

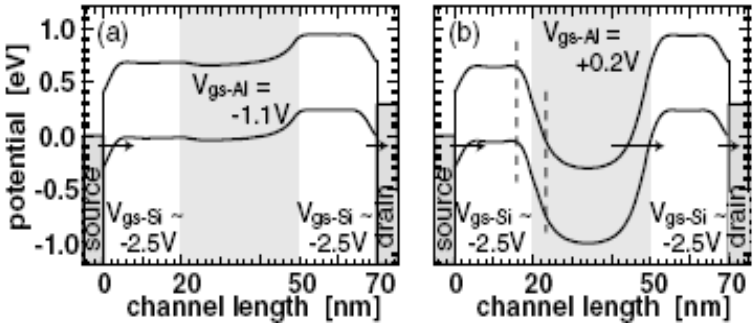
Band-to-band tunneling in CNTs



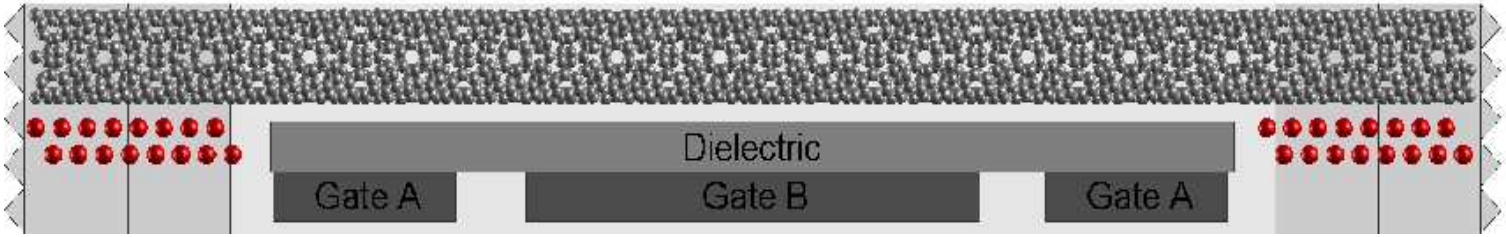


Appenzeller, Lin, Knoch & Avouris
PRL **93**, 196805 (2004)

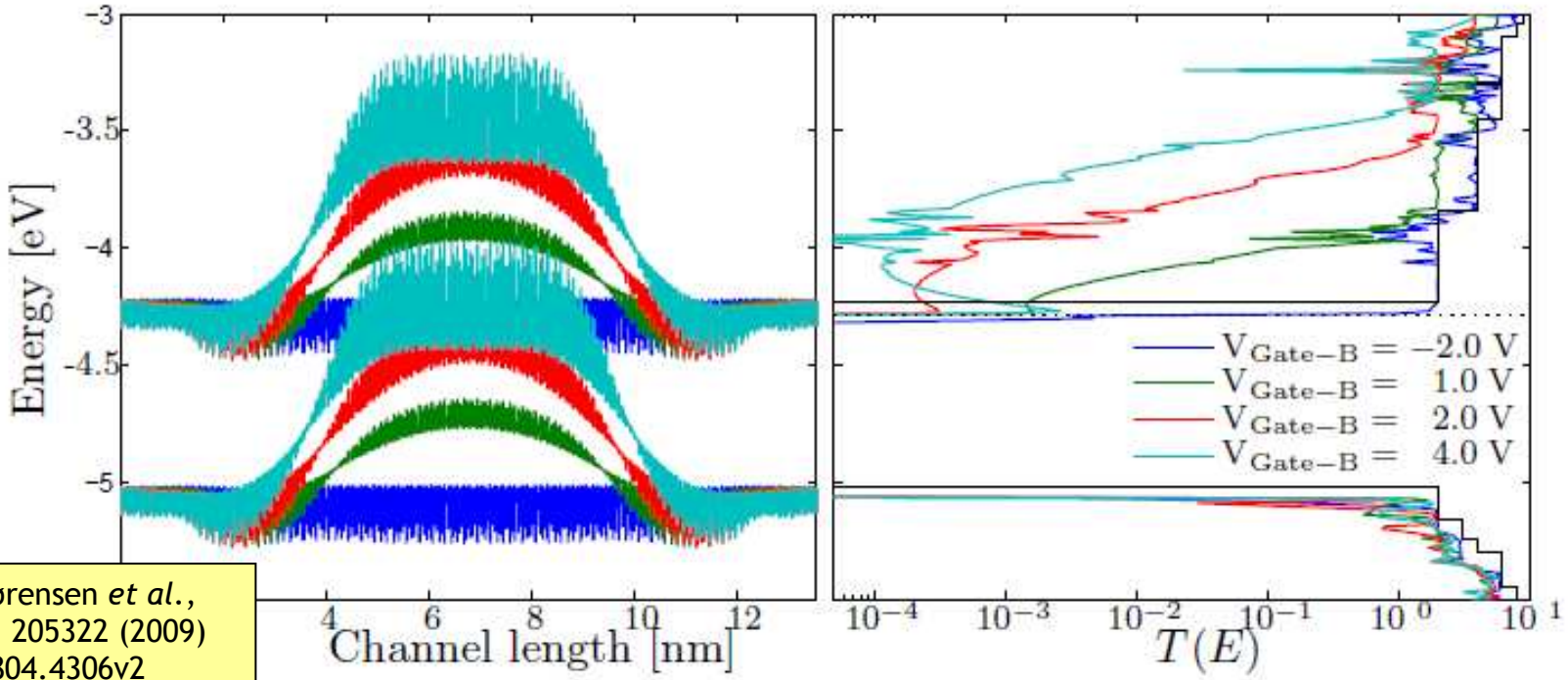




(8,4) nanotube with simple metal electrodes



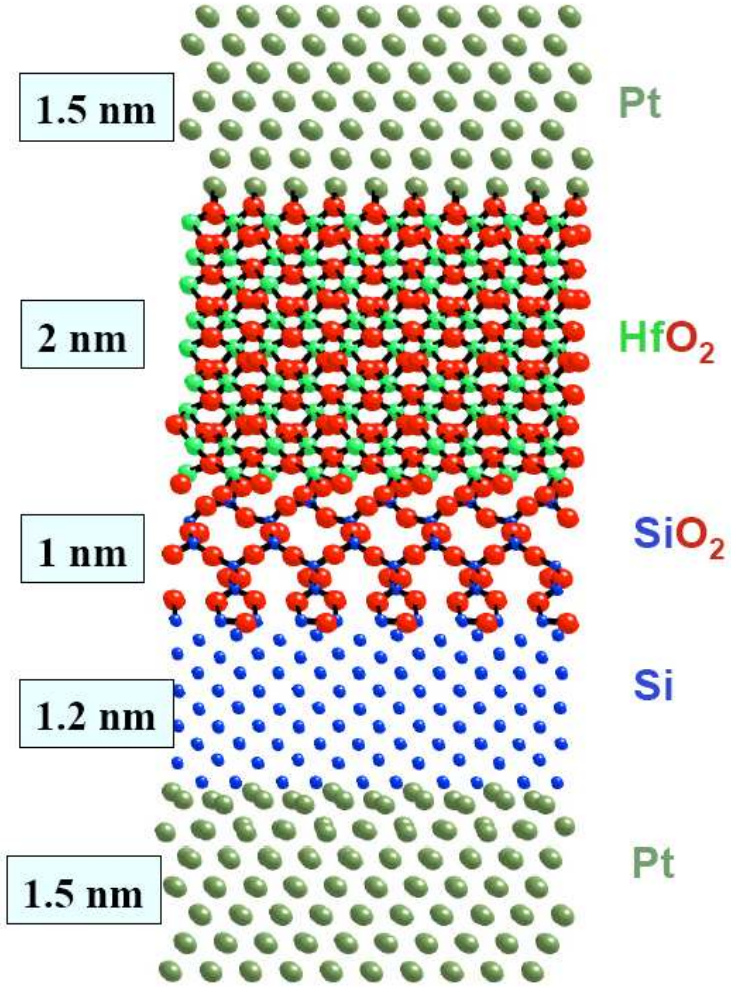
Continuum electrostatic and dielectric regions; 1440 atoms



H. H. Sørensen *et al.*, PRB **79**, 205322 (2009) arXiv:0804.4306v2

High-k dielectrics

- Gate stacks
- DFT can come some of the way
  - » Band gap problems
  - » Calculation time
- Semi-empirical methods can be tuned to high accuracy for specific systems



Grain boundary simulation

Feldman *et al.*, arXiv:0908.2252

- Single grain boundary (GB) reflectivity in Cu and Ag
- Twin/non-twin GB
- Vacancies, disorder, orientation
- Results consistent with experiments
- May point to limitations in the Mayadas-Shatzkes model (one-parameter reflectivity averaged over all GBs)

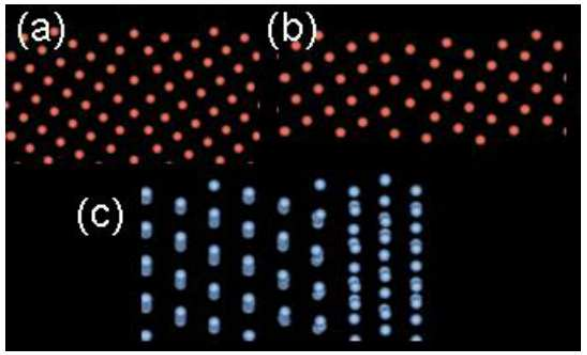
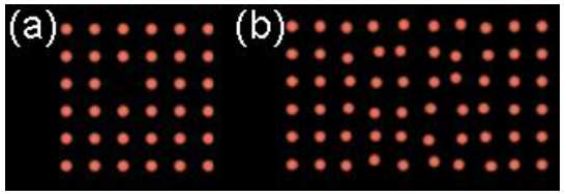


Figure 1. Structure of (a) twin (210) / (120) and (b) twin (320) / (230) boundaries in Cu and (c) non-twin (100) / (110) boundary in Ag. Transmission is from left to right.

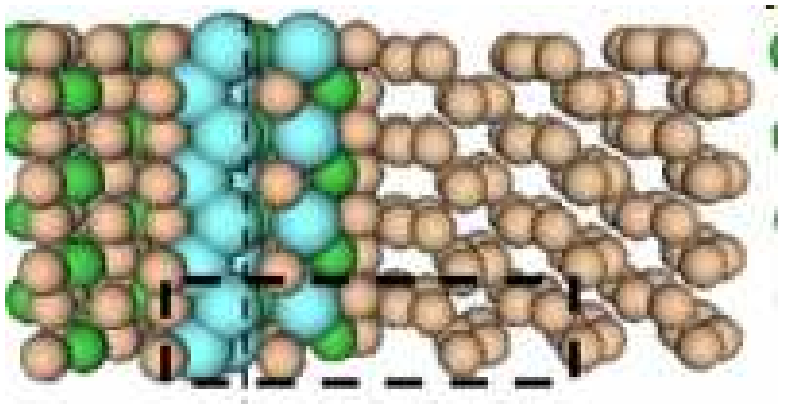


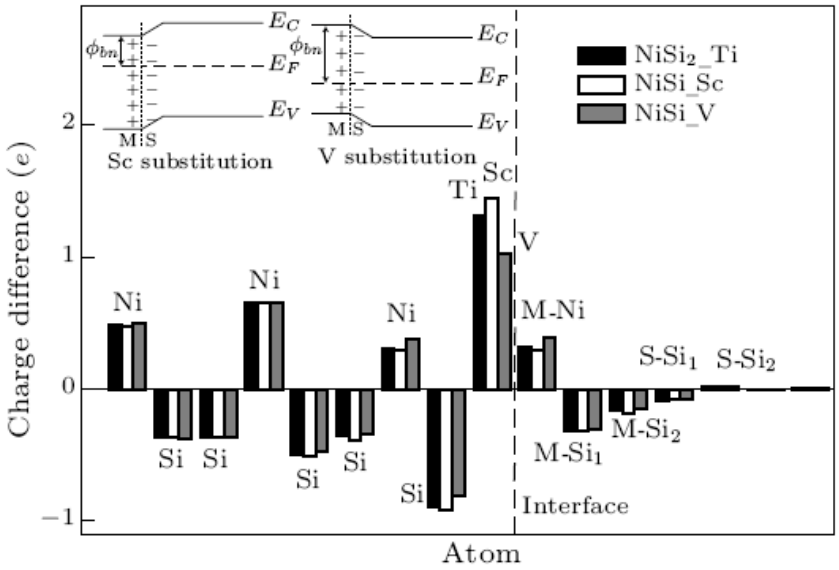


SBH modification

Li Geng *et al.*,
Chinese Physics Letters **26**, 037306 (2009)
IEEE EDL **29**, 746 (2008)

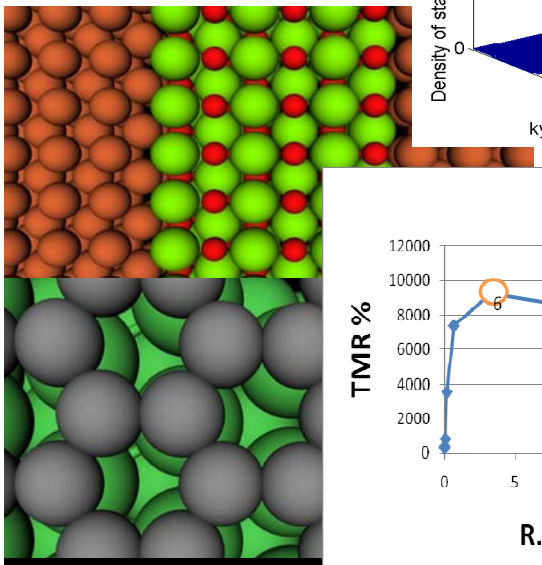
- Fermi level unpinning and Schottky barrier modification by incorporation of Ti, C, V, Y in $NiSi_2$/Si interfaces
- Interface states in gap region are greatly decreased when Si dangling bonds are saturated
- Gives new pinning-free interfacial structure

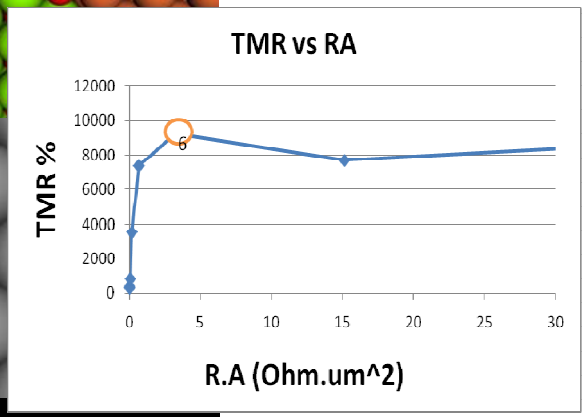


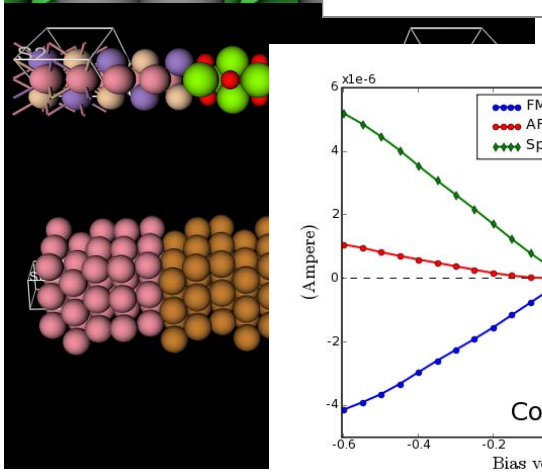


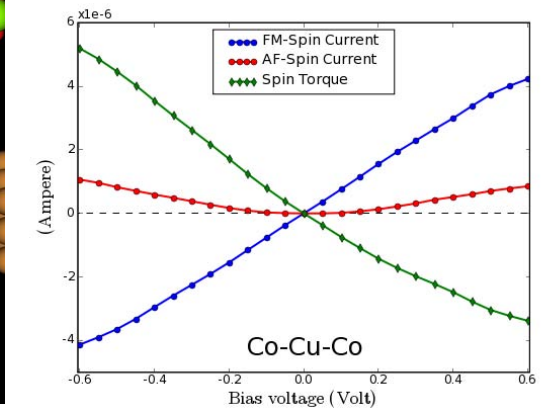
MTJs

- Fe(Co) | MgO | Fe(Co)
- Ni | graphite | Ni
- Heusler alloys
- Co | Cu | Cu

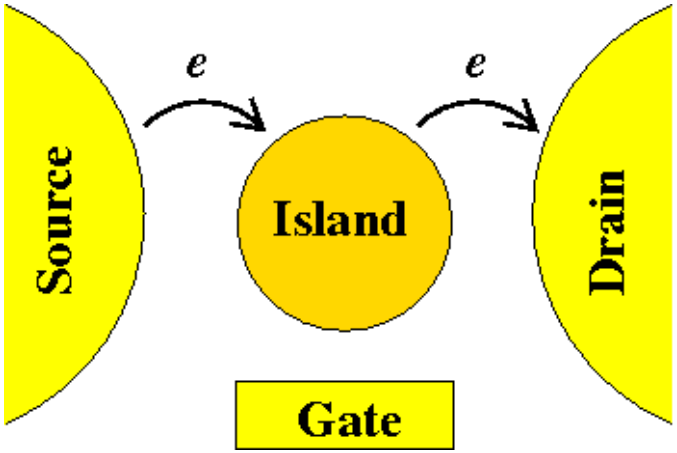




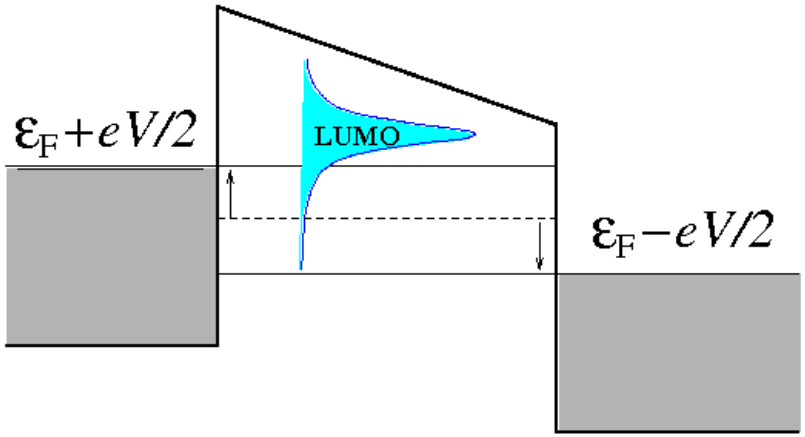




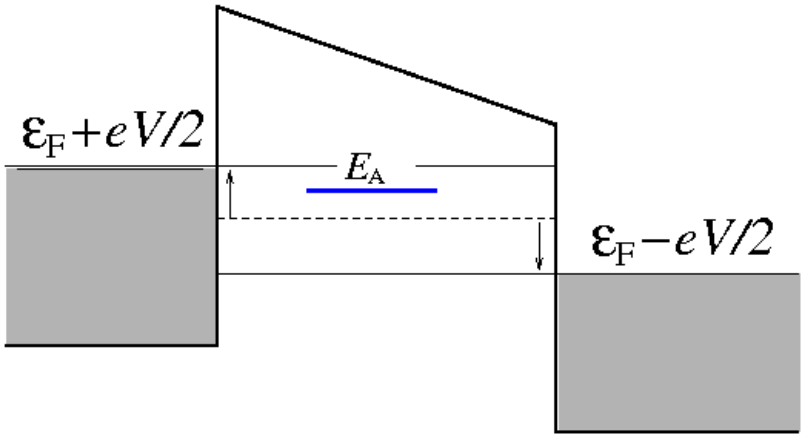
Coulomb blockade transport regime



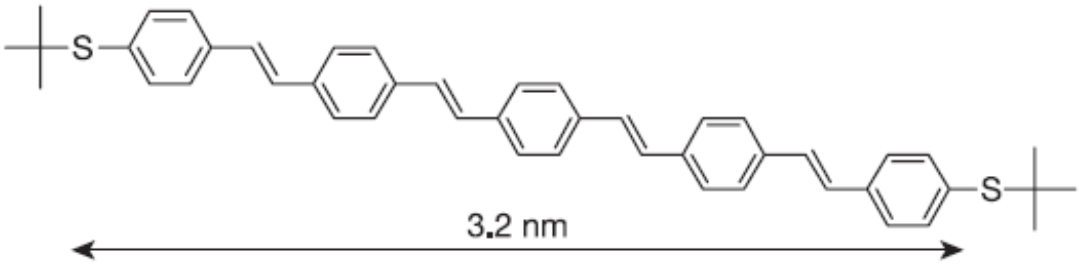
Coherent Tunneling Regime

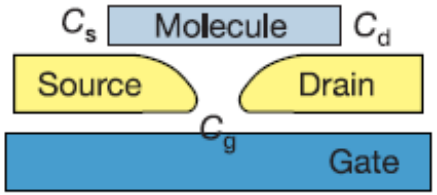


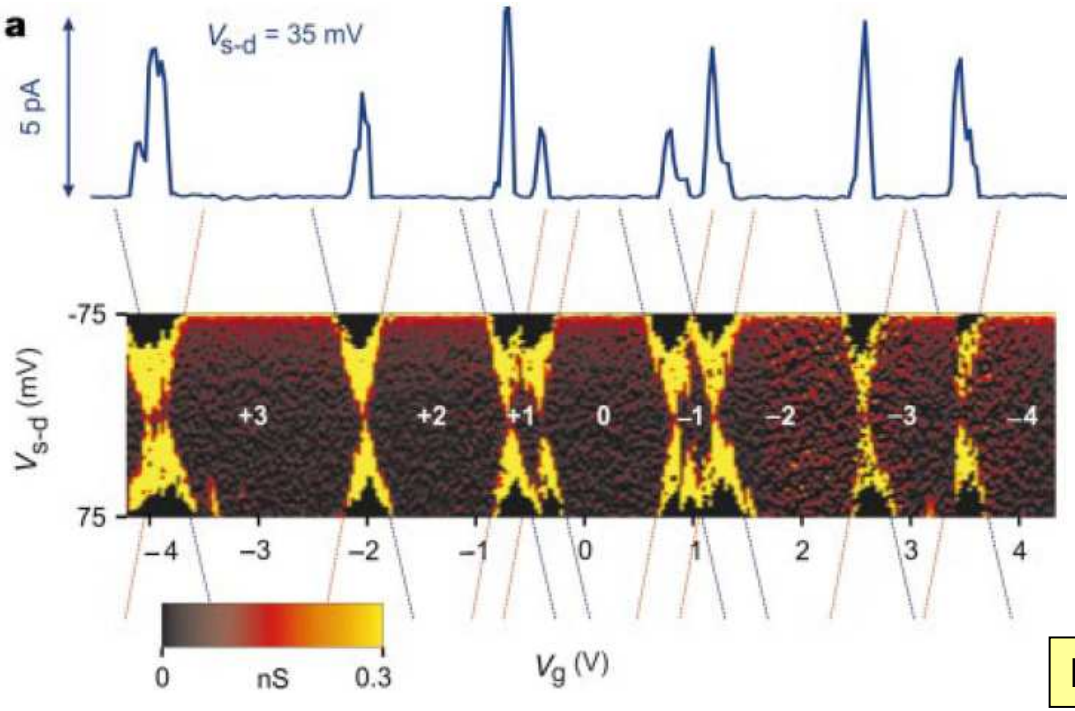
Sequential Tunneling Regime



Excitation gap in molecular SET



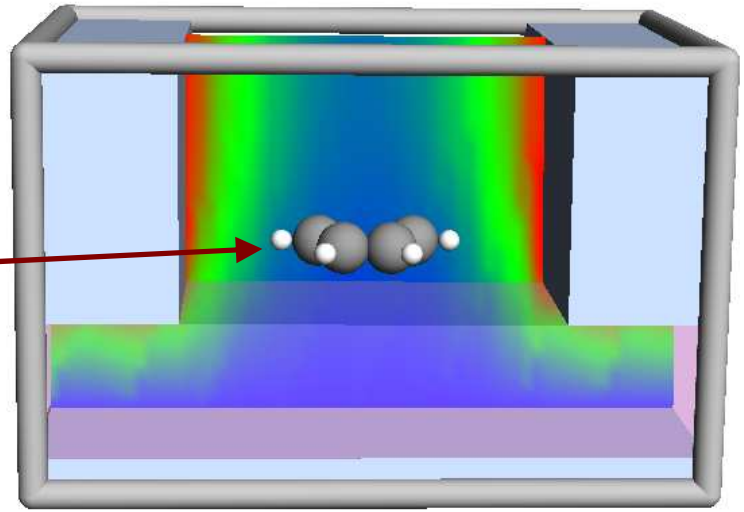




Redox state	Gap (eV) Exp.	Gap (eV) AM1
-3	0,162	
-2	0,262	
-1	0,066	
0	0,230	
1	0,063	0,244
2	0,247	2,188
3	0,358	1,031

Kubatkin *et al.*, *Nature* **425**, 698 (2003)

Charging energy in a SET environment

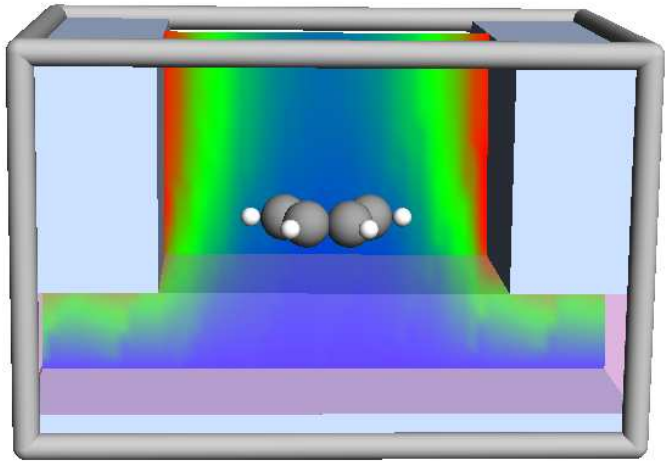


Charged molecule in the gas phase using multi-pole boundary conditions

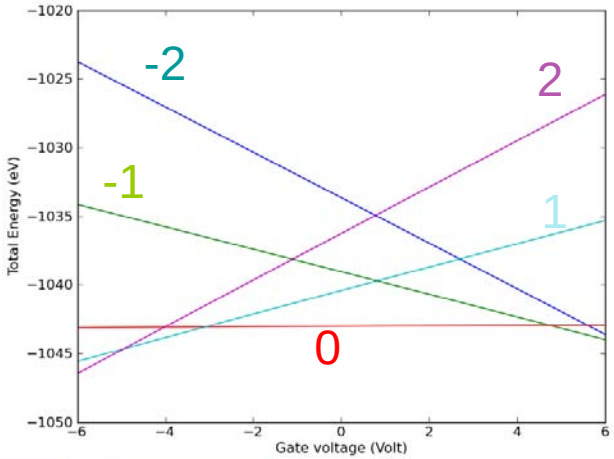
Energies in eV

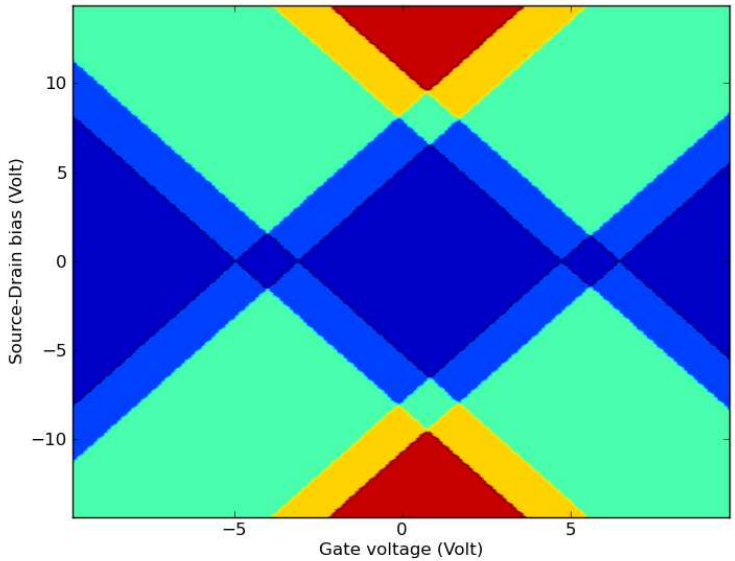
State	SET	Vacuum, neutral	Vacuum, charged	Exp
+2	–9.43		–15.73	
+1	–7.97	–5.90	–9.16	–9.25
0	1.33	0.72	2.35	(>1.29)
–1	–0.11		8.39	

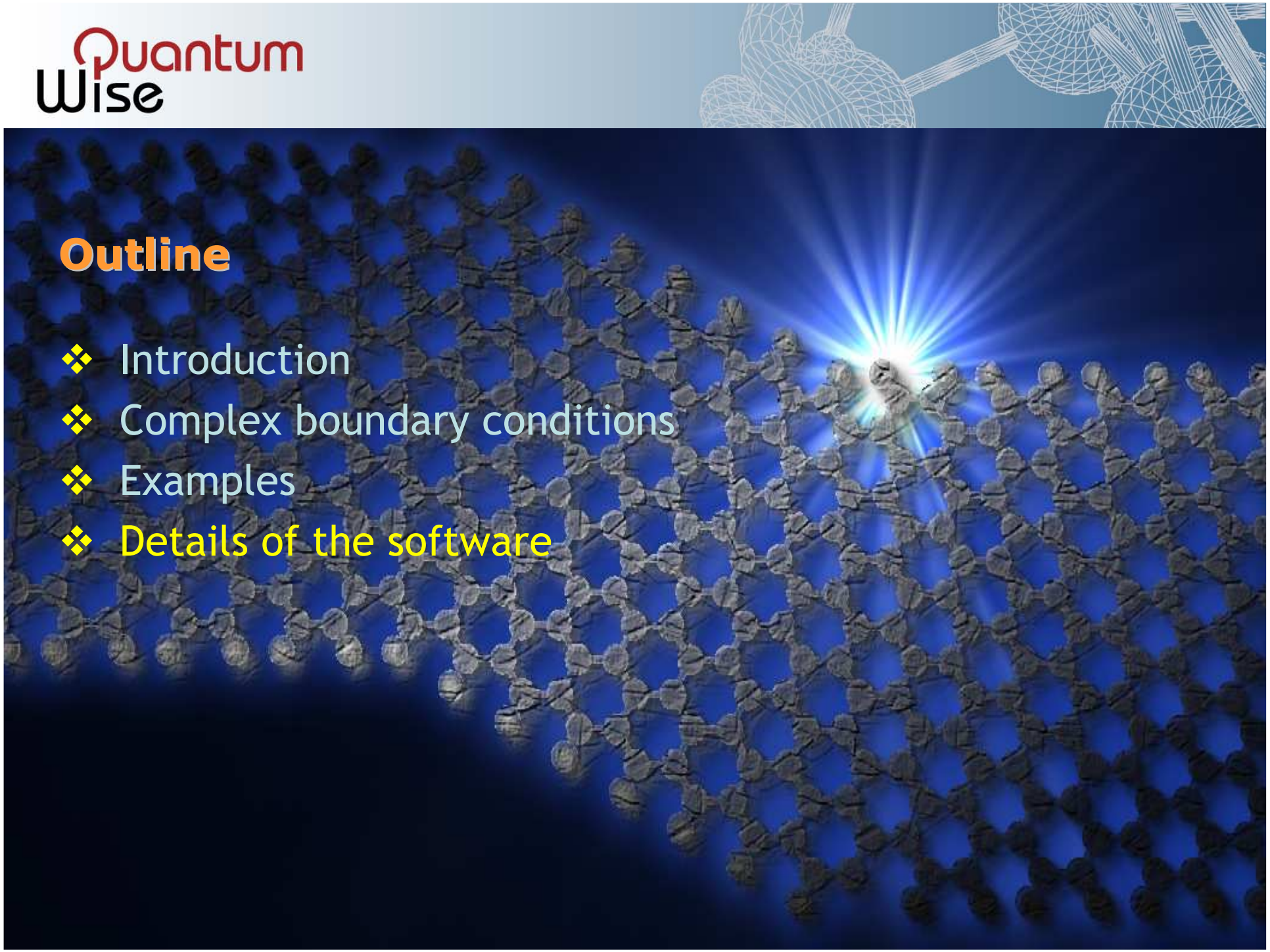
Charge stability diagram



Energy of different charge states

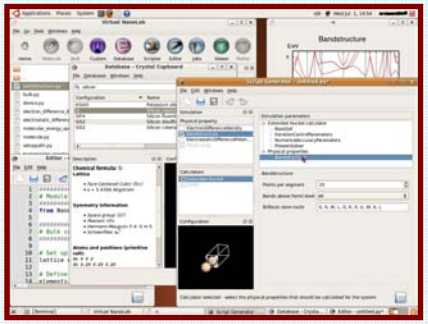





Quantum
Wise
Outline
Introduction
Complex boundary conditions
Examples
Details of the software

Atomistix ToolKit (ATK)

Graphical User Interface



User Scripts

```
from ATK.KohnSham import *
from math import pi,cos,sin

elements = [Oxygen] + 2*[Hydrogen]

z = 0.*Ang
dOH = 0.96*Ang
angle = 104.5*pi/180.
positions = [(z,z,z),
    ( dOH*sin(angle/2), dOH*cos(angle/2), z),
    (-dOH*sin(angle/2), dOH*cos(angle/2), z)]

h2o = MoleculeConfiguration(elements,positions)

calculation = executeSelfConsistentCalculation(h2o,KohnShamMethod())

molecular_spectrum = calculateMolecularEnergySpectrum(calculation)

vnl_file = VNLFile("h2o.vnl")
vnl_file.addToSample(h2o,"Water molecule")
vnl_file.addToSample(molecular_spectrum, "Water molecule")
```

API Modules

Python Interface

Proprietary Engines

- Atomic-scale modeling of electronic devices using DFT or semi-empirical methods
- Massively parallel software
- Fast relaxations
- Integration with continuum modeling

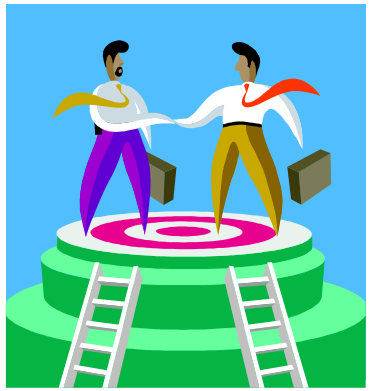
Python Wrappers

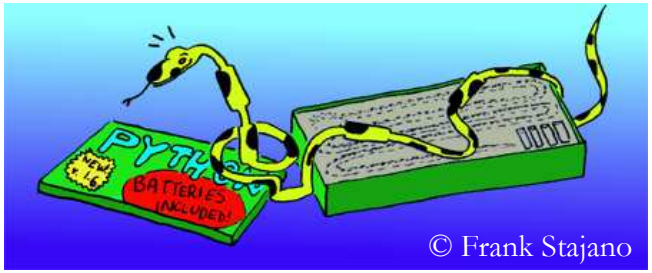
Open-Source / 3rd-Party Software

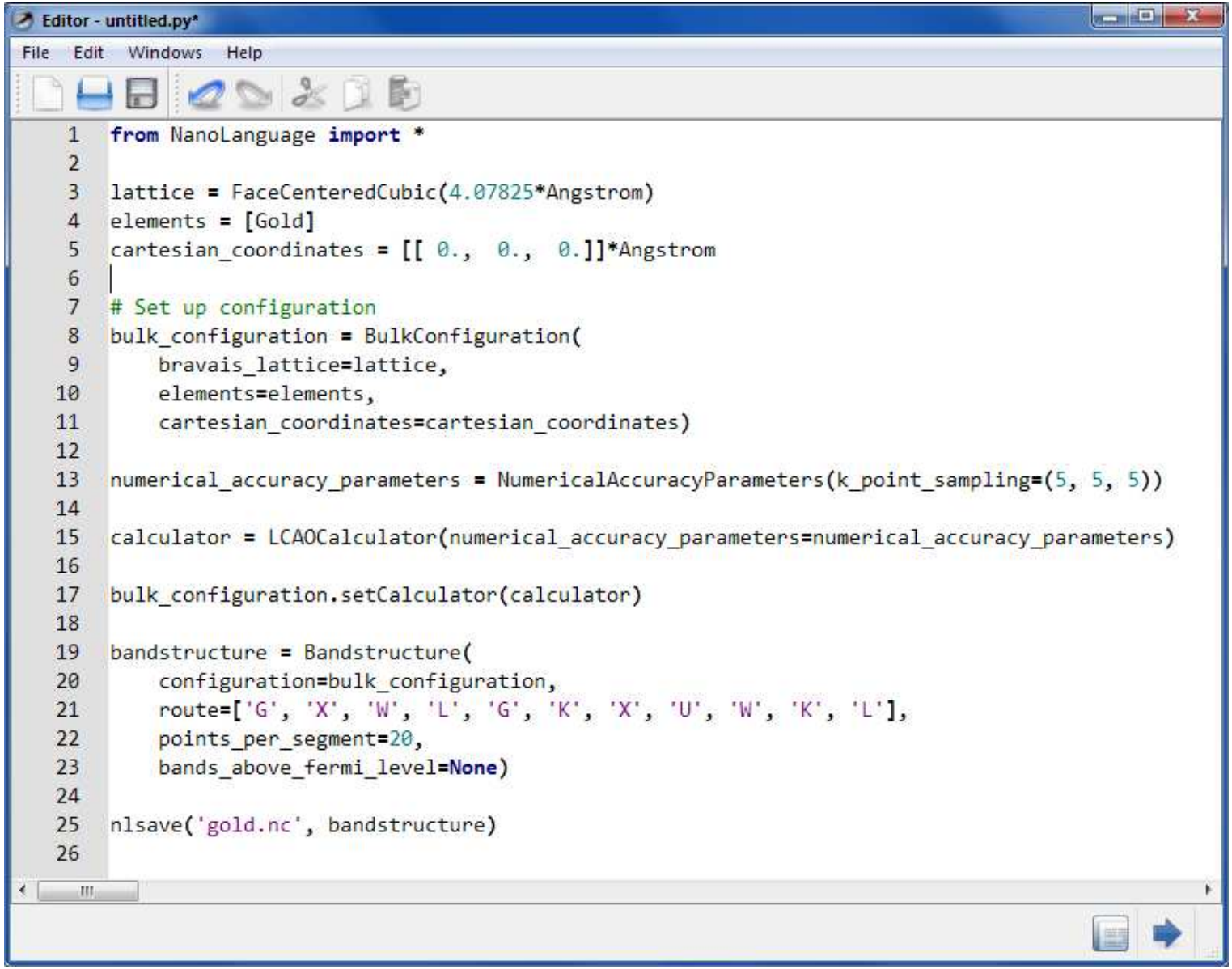
- GPAW, Dacapo
- AbInit, VASP
- Socorro

ATK Python

- **Fully programmable object-oriented scripting interface to ATK**
  - Transparent – the Python syntax is simple yet powerful, with good structural overview
  - Modularized
  - Extendible – integrate your own or third-party algorithms with ATK
- **Tying the platform together**
  - Integrated in the GUI (see later)
  - Wrapping of third-party codes
- **Interpreted language**
  - Can be used interactively
  - Comes with “batteries included” (scientific modules)
  - Performance disadvantage (solvable, all performance critical parts are written in C++, Fortran, or use external optimized libraries)
  - Cross-platform compatible source code

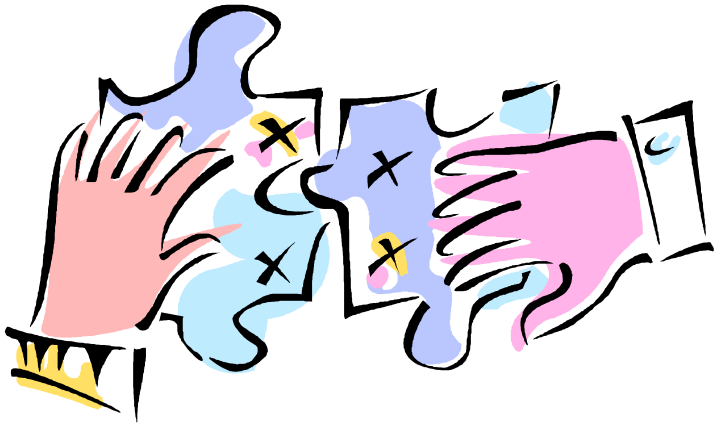





Editor - untitled.py*
File Edit Windows Help
1 from NanoLanguage import *
2
3 lattice = FaceCenteredCubic(4.07825*Angstrom)
4 elements = [Gold]
5 cartesian_coordinates = [[0., 0., 0.]]*Angstrom
6
7 # Set up configuration
8 bulk_configuration = BulkConfiguration(
9   bravais_lattice=lattice,
10   elements=elements,
11   cartesian_coordinates=cartesian_coordinates)
12
13 numerical_accuracy_parameters = NumericalAccuracyParameters(k_point_sampling=(5, 5, 5))
14
15 calculator = LCAOCalculator(numerical_accuracy_parameters=numerical_accuracy_parameters)
16
17 bulk_configuration.setCalculator(calculator)
18
19 bandstructure = Bandstructure(
20   configuration=bulk_configuration,
21   route=['G', 'X', 'W', 'L', 'G', 'K', 'X', 'U', 'W', 'K', 'L'],
22   points_per_segment=20,
23   bands_above_fermi_level=None)
24
25 nlsave('gold.nc', bandstructure)
26

Scripting interface benefits

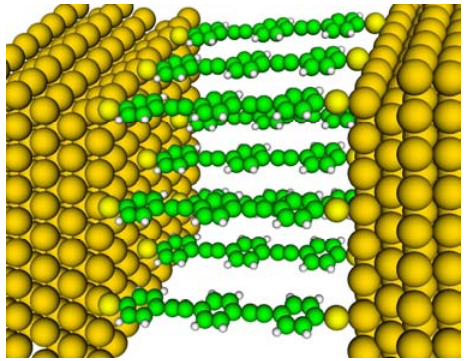
- **Control of parameters**
  - Loop over physical parameters
  - Set up parametrized geometries
  - Detailed control of basis set
- **Control of flow**
  - Converge w.r.t. accuracy parameters
  - Decision-making scripts

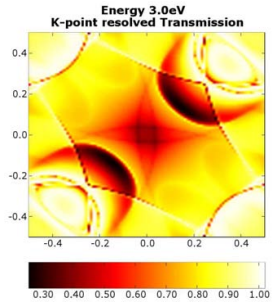


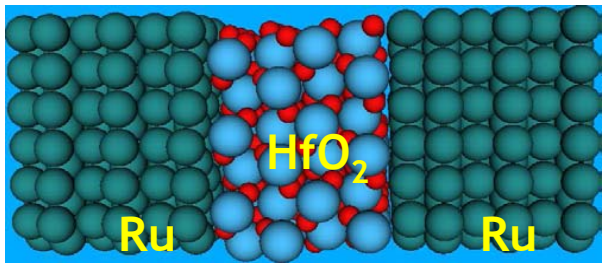
- **Access to raw data**
  - Customized analysis (integrate over spatial coordinates, subtract grids, …)
  - Export data in any desired format
- **Extendibility**
  - Customized modules (3rd party/in-house/from QuantumWise to customer)
  - Implement new algorithms
  - Wrap other software packages
  - Anything you otherwise can do in Python, applied to quantum chemistry and electronic transport!

Graphical user interface (GUI)

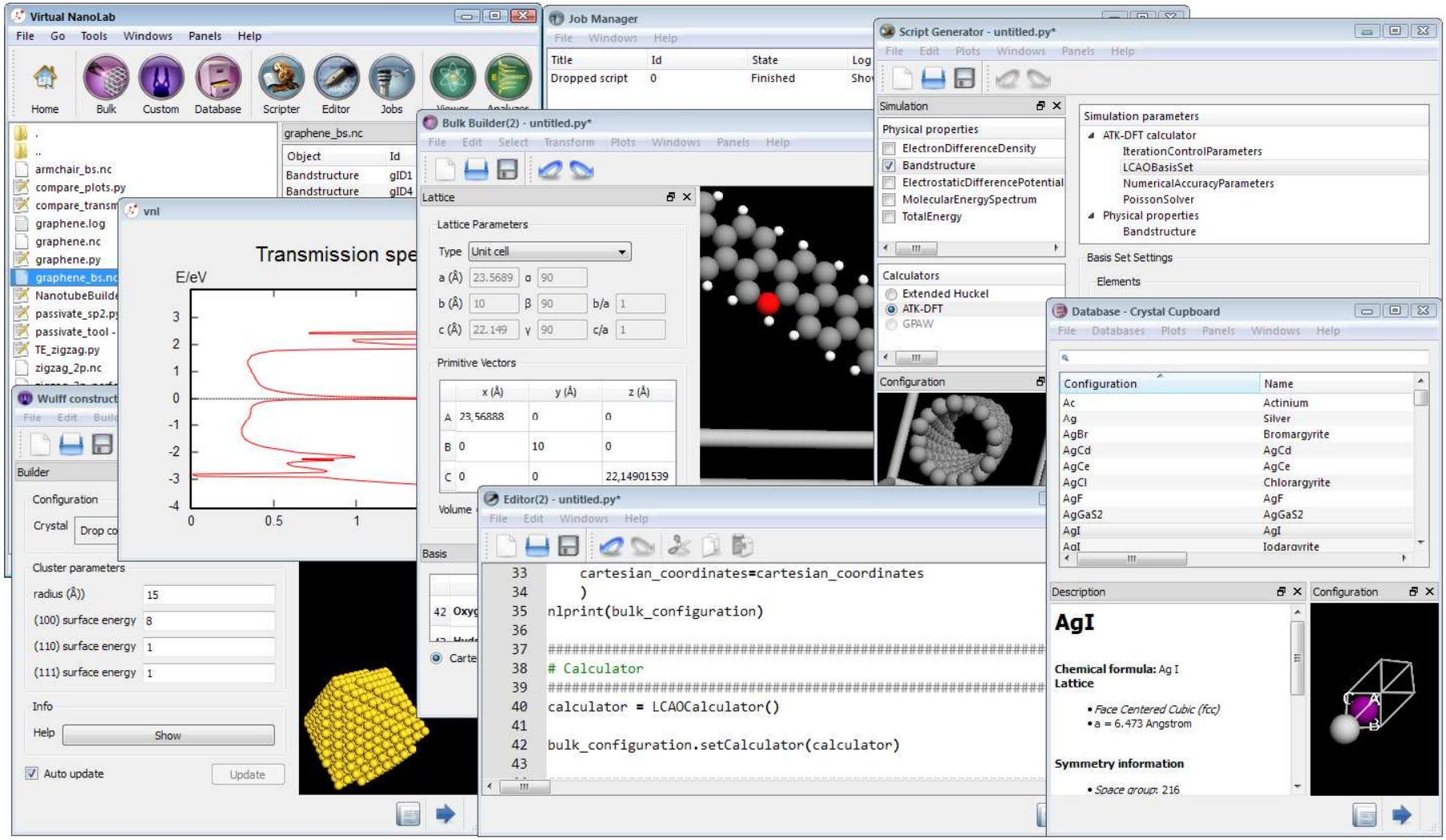
- Guide the user through the work-flow
  - » Set up the device structure
  - » Optimize the geometry
  - » Define the quantum-chemical environment (method & parameters)
  - » Run the calculations
  - » Analyze & plot the results (in 2D and 3D)
- Interactive generation and export of scripts + internal script interpreter
  - » Helps the user learn the command-line interface
  - » Facilitates use of external computational resources
  - » Assists the user during prototyping of scripts
- Plug-ins (both on scripting and GUI level)
  - » User can expand the functionality of the interface
  - » Both geometry setup and analysis modules available
  - » Deliver customized functionality to individual users, or via Forum etc
- Cross-platform compatible input and output files





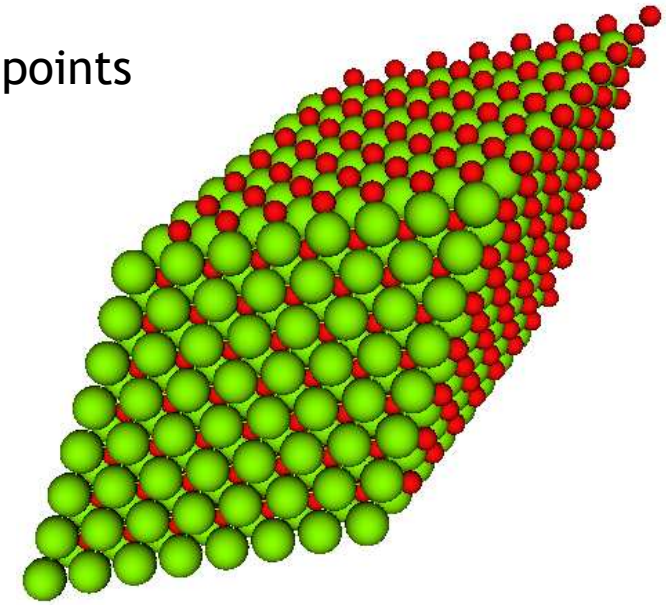


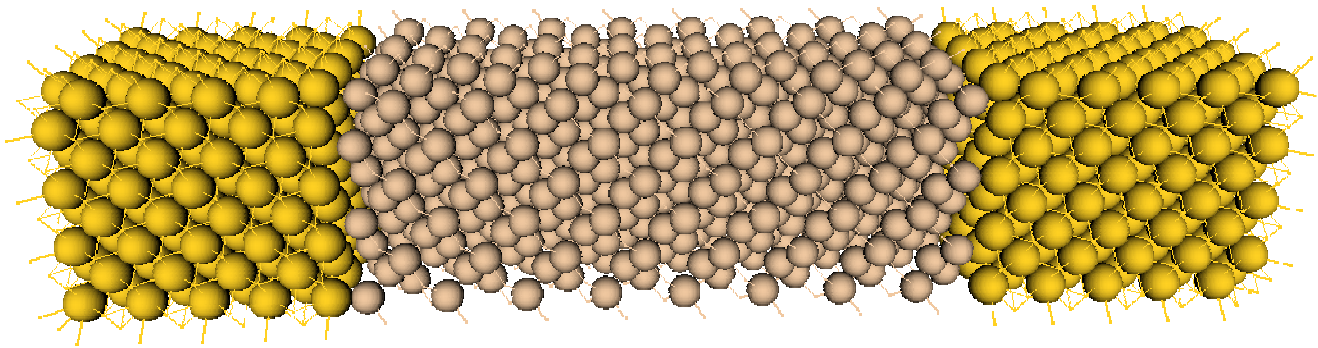
GUI (Virtual NanoLab)



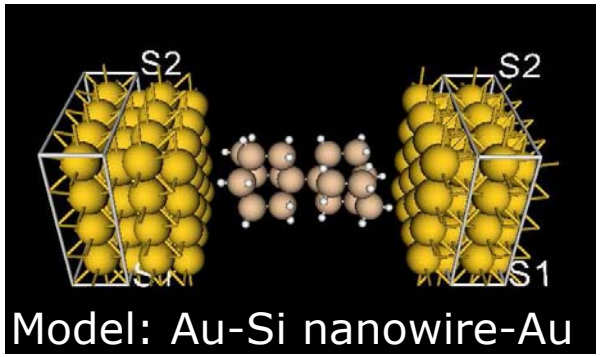
Performance examples

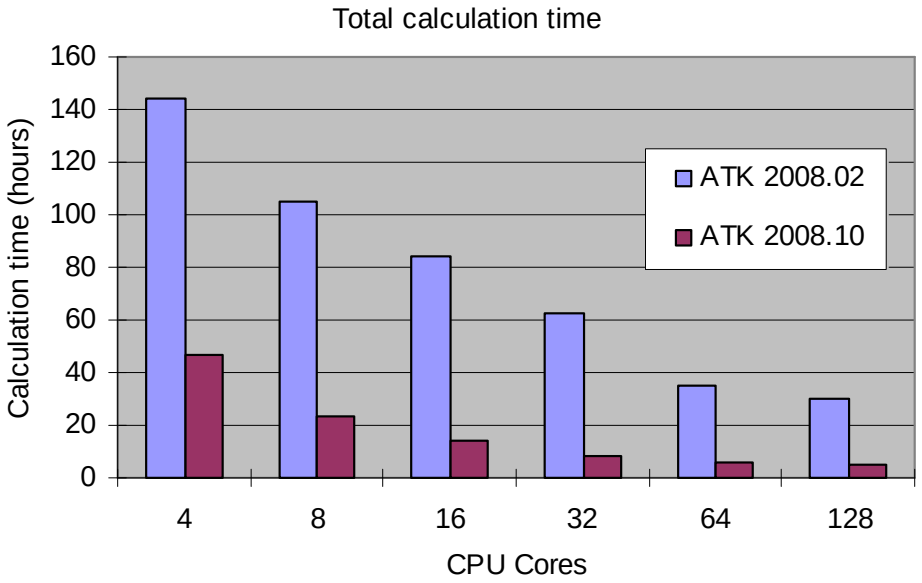
- State-of-the-art parallel DFT code
  - MPI parallelization for k-points and energy points
  - OpenMP for threading on multi-core CPUs
  - Optimized code & libraries provide excellent performance and scaling
- Examples:
  - MgO bulk supercell with 1,024 atoms converges in 16 hours on a single node, using 3 Gb memory (DZP basis set)
  - Au/Si/Au two-probe with 1,166 atoms, 24 hours on 10 nodes, 2 Gb (SZ basis set)

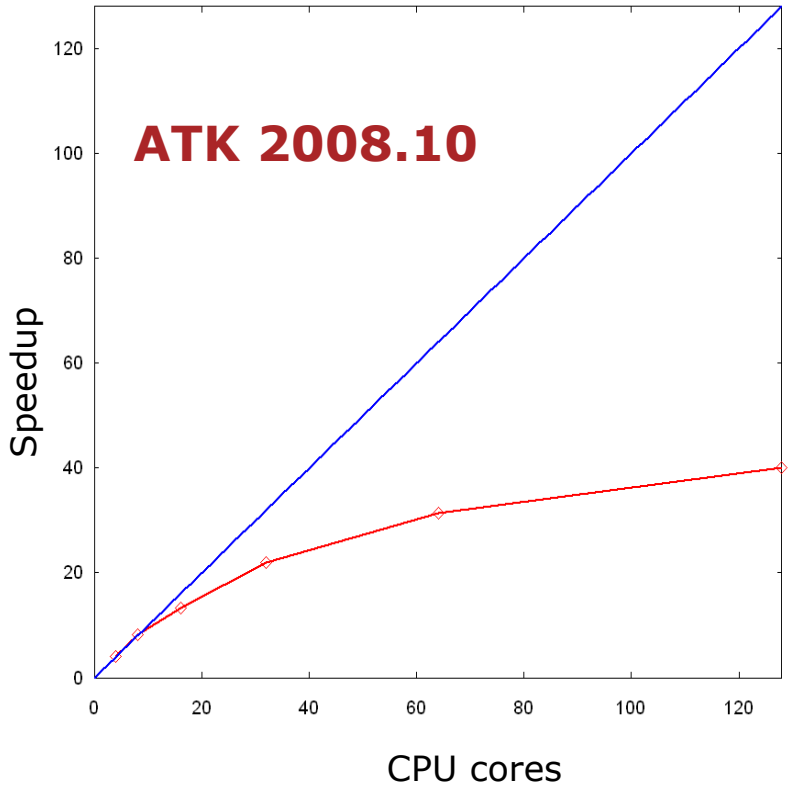




Supercluster parallel scaling & speedup







Summary

- Modeling to complement experiments
- Complex boundary conditions
- Large systems, multi-scale approaches
- Accurate electronic structure
- Response to electrostatic environment
- Modern, user-friendly & flexible software platform

Thank you

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